Beelink GTR9 Pro
UEFI BIOS Setup Reference

Every Setup option exposed by the GTR9 Pro (v2.2) firmware, decoded from the image. Document 1.7.10 · AMD Strix Halo · Ryzen AI Max+ 395 · AMI Aptio V

Board	Beelink GTR9 Pro v2.2 (Realtek RTL8127 NIC)
BIOS image of record	GTRPRPI1001C (32 MB); prior image GTRPR07, identical Setup
SoC	AMD Strix Halo — Ryzen AI Max+ 395, 16C/32T Zen 5, Radeon 8060S (gfx1151)
Power envelope	AMD cTDP 45–120 W; Beelink-validated 140 W on this chassis
Coverage	7 form-sets · 186 forms · 1,010 settings · 988 defaults · 437 performance

Reading the catalog

Columns	Setting · Type · Values. Options are separated by ·; bracketed values are the raw NVRAM values, usable for AMISCE / SCEWIN scripting.
■	Compiled Standard default, printed on the default option.
■ **CHANGE**	Change away from default for a clear gain.
■ **TUNE**	Performance-relevant but workload-specific or expert-only; record originals first.
■ **KEEP**	Default already favors performance; leave it.
Unmarked	No performance dimension — including safety limits and rows inert on this board; leave at default.
Submenus	Every goto prompt on the form, in firmware order. Prompts that open action or confirmation dialogs, runtime-generated forms (BBS boot priorities, Secure Boot key management, RAID dialogs, the firmware-update flow) or driver formsets (Appendix A) have no catalog entry.
Information fields	Read-only text rows (status, versions, sensor values).
Order	Chapters are form-sets, sub-sections are forms, in firmware presentation order.
Elisions	Enumerations of 12 or more collapse to the default plus first/last option and the total option count; a form whose table is byte-identical to an earlier sibling is cross-referenced, not reprinted.
Caveats	Many options are suppress-if / grayout-if hidden, so this is a superset of any one unit's display; live values may differ from compiled defaults. Option and setting strings are reproduced verbatim, firmware typos included.

Platform profile — CachyOS host under ry-install

IOMMU (CBS → NBIO)	Host boots iommu=pt (ry-install 7.164+); keep the firmware default Auto (resolves Enabled) — the NPU, KVM/VFIO passthrough and DMA isolation depend on it. amd_iommu=off is a benchmark-only profile (~6% more iGPU read bandwidth).
UMA Frame buffer Size	Recommendation 512M — RADV and ROCm reach memory through GTT, and the carve-out only hides RAM from the OS. The host runs a 32G carve by choice; from ry-install 7.217.0, GTT stays at the kernel default (~47 GiB, measured 2026-08-14).
Sleep / D3 rows	All systemd sleep targets are masked on this host, so S3 and wake-source rows have no runtime effect; D3 and D3Cold rows still gate device idle at runtime.
NPU (XDNA) rows	amdxdna loads and the NPU is active since ry-install 7.162; NPU-gated options are live.
PCIe link-PM rows	Kernel sets pcie_aspm.policy=performance, which overrides per-port firmware ASPM, L1-substate and Clock PM policy at runtime.
C-states	processor.max_cstate=1 already caps idle depth; CBS Global C-state Control is the firmware-side twin of that choice.
Network Stack / PXE	Disabled in firmware; the OS boots via systemd-boot with no UEFI network path.

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Aptio Setup (Main Menu Tree)

35 forms · 161 settings

Setup (0x2710)

Submenus Main ■ Advanced ■ Chipset ■ Security ■ Boot ■ Save & Exit

Main (0x2711)

Information fields BIOS Vendor, Core Version, Compliancy, BIOS Version, Build Date and Time, Access Level, Serial Number, Total Memory

System Language	Selection	(populated at runtime) [0x0] · (populated at runtime) [0x1]
System Date	Date	Date field (MM/DD/YYYY)
System Time	Time	Time field (HH:MM:SS)

Advanced (0x2712)

Submenus Trusted Computing ■ AMD fTPM configuration ■ ACPI Settings ■ AMD CBS ■ AMD PBS ■ AMD PMF Feature ■ Dash Function ■ Super IO Configuration ■ Hardware Monitor ■ Watch Dog Configuration ■ S5 RTC Wake Settings ■ UEFI Variables Protection ■ CPU Configuration ■ AmiAgesaChipsetWrapper Configuration ■ AMI Graphic Output Protocol Policy ■ Debug Port Table Configuration ■ SIO Common Setting ■ PCI Subsystem Settings ■ USB Configuration ■ Network Stack Configuration ■ NVMe Configuration ■ Demo Board

Trusted Computing (0x2719)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Security Device Support	Selection	Disabled [0x0] · Enabled [0x1] ■

Trusted Computing (0x271A)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Security Device Support	Selection	Disabled [0x0] · Enabled [0x1] ■
TPM State	Selection	Disabled [0x0] · Enabled [0x1] ■
Pending operation	Selection	None [0x0] ■ · TPM Clear [0x5]
TCM State	Selection	Disabled [0x0] · Enabled [0x1] ■
Device Select	Selection	TPM 1.2 [0x0] · TPM 2.0 [0x1] · Auto [0x2] ■

Trusted Computing (0x271B)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Security Device Support	Selection	Disabled [0x0] · Enabled [0x1] ■
SHA-1 PCR Bank	Selection	Disabled [0x0] ■ · Enabled [0x1]
SHA256 PCR Bank	Selection	Disabled [0x0] · Enabled [0x2] ■
SHA384 PCR Bank	Selection	Disabled [0x0] ■ · Enabled [0x4]
SHA512 PCR Bank	Selection	Disabled [0x0] ■ · Enabled [0x8]
SM3_256 PCR Bank	Selection	Disabled [0x0] ■ · Enabled [0x10]
Pending operation	Selection	None [0x0] ■ · TPM Clear [0x1]
Platform Hierarchy	Selection	Disabled [0x0] · Enabled [0x1] ■
Storage Hierarchy	Selection	Disabled [0x0] · Enabled [0x1] ■
Endorsement Hierarchy	Selection	Disabled [0x0] · Enabled [0x1] ■
Physical Presence Spec Version	Selection	1.2 [0x0] · 1.3 [0x1] ■
TPM 2.0 UEFI Spec Version	Selection	TCG_1_2 [0x1] · TCG_2 [0x2] ■
TPM 2.0 InterfaceType	Selection	CRB [0x0] ■ · TIS [0x1]
Device Select	Selection	TPM 1.2 [0x0] · TPM 2.0 [0x1] · Auto [0x2] ■

AMD fTPM configuration (0x271C)

Erase fTPM NV for factory reset	Selection	Disabled [0x0] · Enabled [0x1] ■

ACPI Settings (0x271D)

ACPI Sleep State	Selection	Suspend Disabled [0x0] · S3 (Suspend to RAM) [0x2] ■

Hardware Monitor (0x2735)

Submenus Smart Fan Function

Information fields CPU temperature, System temperature, CPU Fan1 Speed, CPU Fan2 Speed, VCore, V_SM, 3.3V, 5V

Smart Fan Function (0x2737)

CPU Smart Fan1 Mode	Selection	Manual Mode [0x0] · Automatic Mode [0x1] ■ · Full on Mode [0x2] ■ TUNE Automatic Mode (curve) for balance; Full on Mode maximizes cooling/boost headroom at the cost of noise.
Manual PWM Setting	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ ■ TUNE Fixed PWM duty (0–255) used only in Manual Mode; higher means more airflow and thermal headroom at the cost of noise.
Fan off temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x1E ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan start temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x23 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan full speed temperature limit	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x5B ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x46 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] · 0.25 PWM [0x1] · 0.5 PWM [0x2] · 1 PWM [0x3] ■ · 2 PWM [0x4] · 4 PWM [0x5] · 8 PWM [0x6] · 15.875 PWM [0x7] · 4.5 PWM [0x8] ■ TUNE Fan-curve slope (PWM step per degree). A steeper slope ramps airflow faster with temperature for more thermal headroom.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x64 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
CPU Smart Fan2 Mode	Selection	Manual Mode [0x0] · Automatic Mode [0x1] ■ · Full on Mode [0x2] ■ TUNE Automatic Mode for balance, Full on Mode for maximum thermal headroom (louder). Match to Fan1.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] · 0.25 PWM [0x1] · 0.5 PWM [0x2] · 1 PWM [0x3] ■ · 2 PWM [0x4] · 4 PWM [0x5] · 8 PWM [0x6] · 15.875 PWM [0x7] ■ TUNE Fan-curve slope (PWM step per degree). A steeper slope ramps airflow faster with temperature for more thermal headroom.
Smart Fan 3 Mode	Selection	Manual Mode [0x0] ■ · Automatic Mode [0x1] ■ TUNE Automatic Mode for balance; a steeper curve if this fan cools a performance-critical zone.
Fan off temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x10 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan start temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x20 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan full speed temperature limit	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] · 0.25 PWM [0x1] · 0.5 PWM [0x2] ■ · 1 PWM [0x3] · 2 PWM [0x4] · 4 PWM [0x5] · 8 PWM [0x6] · 15.875 PWM [0x7] ■ TUNE Fan-curve slope (PWM step per degree). A steeper slope ramps airflow faster with temperature for more thermal headroom.
Smart Fan 4 Mode	Selection	Manual Mode [0x0] ■ · Automatic Mode [0x1] ■ TUNE Automatic Mode for balance; an aggressive curve holds temperatures down under load.

Watch Dog Configuration (0x2738)

WDT Timeout Mode	Selection	Disabled [0x0] ■ · Minute [0x1] · Second [0x2]
WDT Timeout Value	Numeric	Min 0x1 · Max 0xFF · Step 0x1 · Default 0x1E ■

S5 RTC Wake Settings (0x2739)

Wake system from S5	Selection	Disabled [0x0] ■ · Fixed Time [0x1] · Dynamic Time [0x2]
Wake up hour	Numeric	Min 0x0 · Max 0x17 · Step 0x1 · Default 0x0 ■
Wake up minute	Numeric	Min 0x0 · Max 0x3B · Step 0x1 · Default 0x0 ■
Wake up second	Numeric	Min 0x0 · Max 0x3B · Step 0x1 · Default 0x0 ■
Wake up minute increase	Numeric	Min 0x1 · Max 0x5 · Step 0x1 · Default 0x1 ■

UEFI Variables Protection (0x273A)

Password protection of Runtime Variables	Selection	Enabled [0x1] ■ · Disabled [0x0]

CPU Configuration (0x273C)

Submenus Node 0 Information

PSS Support	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ KEEP Enabled — Exposes ACPI P-states (_PSS) for the acpi-cpufreq fallback; harmless under amd-pstate.
PPC Adjustment	Selection	PState 0 [0x0] ■ ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] · PState 3 [0x3] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] · PState 3 [0x3] · PState 4 [0x4] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] · PState 3 [0x3] · PState 4 [0x4] · PState 5 [0x5] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] · PState 3 [0x3] · PState 4 [0x4] · PState 5 [0x5] · PState 6 [0x6] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
PPC Adjustment	Selection	PState 0 [0x0] ■ · PState 1 [0x1] · PState 2 [0x2] · PState 3 [0x3] · PState 4 [0x4] · PState 5 [0x5] · PState 6 [0x6] · PState 7 [0x7] ■ KEEP PState 0 — _PPC caps the highest core P-state the OS may request; 0 leaves the full boost range available, and higher values only throttle.
NX Mode	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ KEEP Enabled — No-Execute page protection; required by modern OSes at negligible cost.

AmiAgesaChipsetWrapper Configuration (0x2741)

PCR7 WA Support	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x2] ■
Enable Dual Vga Controllers	Selection	Disabled [0x0] ■ · Enabled [0x1]

AMI Graphic Output Protocol Policy (0x274B)

Output Select	Selection	Unknown Device [0x0] ■
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1]
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2]
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2] · Unknown Device [0x3]
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2] · Unknown Device [0x3] · Unknown Device [0x4]

Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2] · Unknown Device [0x3] · Unknown Device [0x4] · Unknown Device [0x5]
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2] · Unknown Device [0x3] · Unknown Device [0x4] · Unknown Device [0x5] · Unknown Device [0x6]
Output Select	Selection	Unknown Device [0x0] ■ · Unknown Device [0x1] · Unknown Device [0x2] · Unknown Device [0x3] · Unknown Device [0x4] · Unknown Device [0x5] · Unknown Device [0x6] · Unknown Device [0x7]

Debug Port Table (0x274D)

Debug Port Table	Selection	Disabled [0x0] ■ · Enabled [0x1]
Debug Port Table 2	Selection	Disabled [0x0] ■ · Enabled [0x1]

USB Configuration (0x2750)

Information fields USB Module Version, USB Controllers:, USB Devices:, USB hardware delays and time-outs:, Mass Storage Devices:

USB Support	Selection	Disabled [0x0] · Enabled [0x1] ■
Legacy USB Support	Selection	Enabled [0x0] · Disabled [0x1] ■ · Auto [0x2]
XHCI Legacy Support	Selection	Enabled [0x1] ■ · Disabled [0x0]
XHCI Hand-off	Selection	Enabled [0x1] ■ · Disabled [0x0]
EHCI Hand-off	Selection	Disabled [0x0] ■ · Enabled [0x1]
USB Mass Storage Driver Support	Selection	Disabled [0x0] · Enabled [0x1] ■
USB transfer time-out	Selection	1 sec [0x1] · 5 sec [0x5] · 10 sec [0xA] · 20 sec [0x14] ■
Device reset time-out	Selection	10 sec [0x0] · 20 sec [0x1] ■ · 30 sec [0x2] · 40 sec [0x3]
Device power-up delay	Selection	Auto [0x0] ■ · Manual [0x1]
Device power-up delay in seconds	Numeric	Min 0x1 · Max 0x28 · Step 0x1 · Default 0x5 ■
Mass Storage Device Emulation Type	Selection	Per-device emulation override, repeated once per detected USB mass-storage device. Options (identical for each): Auto [0x0] ■ · Floppy [0x1] · Forced FDD [0x2] · Hard Disk [0x3] · CD-ROM [0x4]

Network Stack Configuration (0x2752)

Network Stack	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ KEEP Disabled — Removes PXE/IPv4/IPv6 UEFI drivers from the boot path.
IPv4 PXE Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
IPv4 HTTP Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
IPv6 PXE Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
IPv6 HTTP Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
PXE boot wait time	Numeric	Min 0x0 · Max 0x5 · Step 0x1 · Default 0x0 ■
Media detect count	Numeric	Min 0x1 · Max 0x32 · Step 0x1 · Default 0x1 ■

Demo Board (0x2964)

Submenus PCI-E Port

Onboard PCIE LAN PXE ROM	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Disabled shortens POST when no network boot is used.

PCI-E Port (0x2966)

Pcie Port Control	Selection	Disabled [0x0] · Enabled [0x1] ■
Device0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device0)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device0)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device1	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device1)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device1)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device2	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■

ASPM Mode Control(Device2)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device2)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device3	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device3)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device3)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device4	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device4)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device4)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device5	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device5)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device5)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device6	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device6)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device6)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■
Device7	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
ASPM Mode Control(Device7)	Selection	Disabled [0x0] · L0s Entry [0x1] · L1 Entry [0x2] · L0s And L1 Entry [0x3] · Auto [0xFF] ■ ■ TUNE PCIe link power management. Disabled means no L0s/L1 entry and the lowest latency (best performance, higher idle power).
Hotplug Mode Control(Device7)	Selection	Disabled [0x0] · Hotplug Basic [0x1] · Hotplug Server [0x2] · Hotplug Enhanced [0x3] · Hotplug Inboard [0x4] · Auto [0xFF] ■

Chipset (0x2713)

Submenus South Bridge ■ GFX Configuration ■ North Bridge

South Bridge (0x2725)

Submenus SB USB Configuration ■ SB MISC DEBUG Configuration ■ 17113 (string token 0x42D9, no text)

SB USB Configuration (0x2726)

Submenus USB0 Ports ■ USB1 Ports ■ USB3 Ports ■ USB4 Ports

USB0 Ports (0x2727)

XHCI0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 Port 1	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 2.0 Port 1	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 2.0 Port 2	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 2.0 Port 3	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI0 2.0 Port 4	Selection	Disabled [0x0] · Enabled [0x1] ■

USB1 Ports (0x2728)

XHCI1 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI1 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■

USB3 Ports (0x2729)

XHCI3 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI3 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■

USB4 Ports (0x272A)

XHCI4 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■
XHCI4 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] ■

SB MISC DEBUG Configuration (0x272B)

HPET ACPI table In SB	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Enabled (Auto) — Modern AMD platforms use invariant TSC; forcing HPET off rarely helps and can break timing-sensitive software.

North Bridge (0x273F)

Submenus Socket 0 Information

Security (0x2714)

Submenus Secure Boot ■ Secure Flash Update

Information fields Minimum length, Maximum length

Administrator Password	Password	Password entry
User Password	Password	Password entry

Secure Boot (0x2815)

Submenus Restore Factory Keys ■ Reset To Setup Mode ■ Expert Key Management

Information fields System Mode

Secure Boot	Selection	Disabled [0x0] ■ · Enabled [0x1]
Secure Boot Mode	Selection	Standard [0x0] · Custom [0x1] ■

Expert Key Management (0x281F)

Submenus Restore Factory Keys ■ Reset To Setup Mode ■ Enroll Efi Image ■ Export Secure Boot variables

Information fields Vendor Keys

Factory Key Provision	Selection	Disabled [0x0] ■ · Enabled [0x1]

Secure Flash Update (0x282E)

Information fields Signed FW Update, FW Key File, FW Key Type, FW Update Methods, FW Revision Rollback, Flash Write Protect

Boot (0x2715)

Submenus UEFI NVME Drive BBS Priorities ■ UEFI Hard Disk Drive BBS Priorities ■ UEFI CDROM/DVD Drive BBS Priorities ■ UEFI SD Drive BBS Priorities ■ UEFI USB Drive BBS Priorities ■ UEFI NETWORK Drive BBS Priorities ■ UEFI Application Boot Priorities

Setup Prompt Timeout	Numeric	Min 0x1 · Max 0xFFFF · Step 0x1 · Default 0x1 ■
Bootup NumLock State	Selection	On [0x1] ■ · Off [0x0]
Fast Boot	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Enabled for faster POST (skips some device initialization). Disable temporarily if you need full enumeration or to enter Setup reliably.
SATA Support	Selection	Last Boot SATA Devices Only [0x0] ■ · All SATA Devices [0x1]
NVMe Support	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ KEEP Enabled — Required to boot from NVMe; no downside.
UFS Support	Selection	Disabled [0x0] · Enabled [0x1] ■
VGA Support	Selection	Auto [0x0] ■ · EFI Driver [0x1]
USB Support	Selection	Disabled [0x0] · Full Initial [0x1] ■ · Partial Initial [0x2]
PS2 Devices Support	Selection	Disabled [0x0] · Enabled [0x1] ■
Network Stack Driver Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Redirection Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Driver Option #%d	Selection	(populated at runtime) [0x0] ■ · (populated at runtime) [0x1]
Boot mode select	Selection	LEGACY [0x0] · UEFI [0x1] ■ ■ KEEP UEFI (not Legacy/CSM) — UEFI is required for Resizable BAR and modern features; CSM blocks Above-4G/ReBAR.
Boot Option #1	Selection	NVME [0x0] ■ · Hard Disk [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] · UEFI AP [0x6] · Disabled [0x7]
Boot Option #2	Selection	NVME [0x0] · Hard Disk [0x1] ■ · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] · UEFI AP [0x6] · Disabled [0x7]
Boot Option #3	Selection	NVME [0x0] · Hard Disk [0x1] · CD/DVD [0x2] ■ · SD [0x3] · USB Device [0x4] · Network [0x5] · UEFI AP [0x6] · Disabled [0x7]
Boot Option #4	Selection	NVME [0x0] · Hard Disk [0x1] · CD/DVD [0x2] · SD [0x3] ■ · USB Device [0x4] · Network [0x5] · UEFI AP [0x6] · Disabled [0x7]

Boot Option #5	Selection	NVME [0x0] · Hard Disk [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] ■ · Network [0x5] · UEFI AP [0x6] · Disabled [0x7]
Boot Option #6	Selection	NVME [0x0] · Hard Disk [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] ■ · UEFI AP [0x6] · Disabled [0x7]
Boot Option #7	Selection	NVME [0x0] · Hard Disk [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] · UEFI AP [0x6] ■ · Disabled [0x7]
Boot Option #1	Selection	Hard Disk [0x0] ■ · NVME [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] · Disabled [0x6]
Boot Option #2	Selection	Hard Disk [0x0] · NVME [0x1] ■ · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] · Disabled [0x6]
Boot Option #3	Selection	Hard Disk [0x0] · NVME [0x1] · CD/DVD [0x2] ■ · SD [0x3] · USB Device [0x4] · Network [0x5] · Disabled [0x6]
Boot Option #4	Selection	Hard Disk [0x0] · NVME [0x1] · CD/DVD [0x2] · SD [0x3] ■ · USB Device [0x4] · Network [0x5] · Disabled [0x6]
Boot Option #5	Selection	Hard Disk [0x0] · NVME [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] ■ · Network [0x5] · Disabled [0x6]
Boot Option #6	Selection	Hard Disk [0x0] · NVME [0x1] · CD/DVD [0x2] · SD [0x3] · USB Device [0x4] · Network [0x5] ■ · Disabled [0x6]

AMD CBS (Common BIOS Settings)

51 forms · 318 settings

AMD CBS (0x7000)

Submenus CPU Common Options ■ DF Common Options ■ UMC Common Options ■ NBIO Common Options ■ FCH Common Options ■ SMU Common Options ■ SOC Miscellaneous Control

Information fields AMD CBS Revision Number

Combo CBS	Numeric	Min 0x0 · Max 0xFF · Default 0xFE ■

CPU Common Options (0x7001)

Submenus Performance ■ Prefetcher settings ■ Core Watchdog

REP-MOV/STOS Streaming	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ KEEP Enabled — Streaming string ops improve large memcpy/memset bandwidth.
RedirectForReturnDis	Selection	Auto [0xFF] ■ · 1 [0x1] · 0 [0x0]
Platform First Error Handling	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0x3] ■
Core Performance Boost	Selection	Disabled [0x0] · Auto [0x1] ■ ■ KEEP Auto — Disabling forfeits all boost above base clock.
Global C-state Control	Selection	Disabled [0x0] ■ · Enabled [0x1] · Auto [0x3] ■ TUNE Enabled for idle power; Disabled trims wake latency. Redundant if the kernel already caps C-states (processor.max_cstate=1).
Opcache grayout flag	Selection	0 [0x0] ■ · 1 [0x1] · Display [0x2]
Opcache Control	Selection	Disabled [0x1] · Enabled [0x0] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — The decoded-op cache improves front-end throughput and efficiency; leave it on.
SEV ASID Count	Selection	253 ASIDs [0x0] · 509 ASIDs [0x1] · Auto [0x3] ■
SEV-ES ASID Space Limit Control	Selection	Auto [0x0] ■ · Manual [0x1]
SEV-ES ASID Space Limit	Numeric	Min 0x1 · Max 0x1FE · Default 0x1 ■
Streaming Stores Control	Selection	Disabled [0x1] · Enabled [0x0] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — Non-temporal stores help bandwidth-bound workloads.
Local APIC Mode	Selection	Compatibility [0x0] · xAPIC [0x1] · x2APIC [0x2] · Auto [0xFF] ■ ■ KEEP x2APIC (Auto) — x2APIC improves interrupt scaling on high-core-count parts; Auto selects correctly.
ACPI _CST C1 Declaration	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■ ■ KEEP Enabled (Auto) — Lets the OS use C1 for efficiency without measurable throughput loss.
MCA error thresh enable	Selection	False [0x0] · True [0x1] · Auto [0xFF] ■
MCA error thresh count	Numeric	Min 0x1 · Max 0xFFF · Default 0xFF5 ■
SMU and PSP Debug Mode	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■
Enhanced REP MOVSB/STOSB (ERSM)	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — Faster string-copy microcode path.
Log Transparent Errors	Selection	Auto [0x3] ■ · Disabled [0x0] · Enabled [0x1]
AVX512	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — AVX-512 accelerates vectorized/AI workloads on Zen 5; only disable for thermal-limited edge cases.
MONITOR and MWAIT disable	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xFF] ■ ■ KEEP Auto (MWAIT stays enabled) — MWAIT-based idle is more efficient and lower-latency than polling.
SVM Lock	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xFF] ■
SVM Enable	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — Required for KVM. Pair with IOMMU Enabled for VFIO passthrough.
Corrector Branch Predictor	Selection	Disable [0x0] ■ · Enable [0x1] ■ TUNE Enable — Corrector predictor stage; the compiled default is Disable.
PAUSE Delay	Selection	Auto [0xFF] ■ · Disable [0x0] · 16 cycles [0x1] · 32 cycles [0x2] · 64 cycles [0x3] · 128 cycles [0x4]

Performance (0x7008)

Submenus Custom Core Pstates

OC Mode	Selection	Normal Operation [0x0] · OC1 [0x1] · OC2 [0x2] · OC3 [0x3] · Customized [0x5] ■ ■ TUNE Normal Operation unless you have a deliberate tuning plan; the compiled default is Customized, which unlocks manual P-states and is expert-only.

Custom Core Pstates (0x700F)

Submenus Decline ■ Accept

Information fields AMD out-of-specification operation warning — Decline / Accept gate; damage from out-of-spec use is not covered by warranty.

Accept (0x7011)

Custom Pstate0	Selection	Custom [0x1] · Auto [0x2] ■ ■ TUNE Auto — Expert-only manual P0 definition; leave the default unless you know your stable frequency/VID pair.
Pstate0 Freq (MHz)	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Manual P0 frequency; use only a validated frequency/VID pair. PBO + CO is safer for gains.
Pstate0 VID	Numeric	Min 0x0 · Max 0x1FF · Default 0x0 ■ ■ TUNE Manual P0 voltage ID; pair it carefully with the frequency, since wrong values prevent boot.

Prefetcher settings (0x7009)

L1 Stream HW Prefetcher	Selection	Disable [0x0] · Enable [0x1] · Auto [0x3] ■ ■ KEEP Enable (Auto) — Hardware prefetch hides memory latency; broadly positive. Disable only if profiling a specific regression.
L2 Stream HW Prefetcher	Selection	Disable [0x0] · Enable [0x1] · Auto [0x3] ■ ■ KEEP Enable (Auto) — Improves memory-bound throughput; leave it on.
L2 Up/Down Prefetcher	Selection	Disable [0x0] · Enable [0x1] · Auto [0x3] ■ ■ KEEP Enable (Auto) — Bidirectional stride prefetch; generally beneficial.

Core Watchdog (0x700A)

Core Watchdog Timer Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■
Core Watchdog Timer Interval	Selection	Auto [0xFFFF] ■ · 21 options: 21.461s [0x900] … 39.68us [0x701]
Core Watchdog Timer Severity	Selection	No Error [0x0] · Transparent [0x1] · Corrected [0x2] · Deferred [0x3] · Uncorrected [0x4] · Fatal [0x5] · Auto [0xFF] ■

DF Common Options (0x7002)

Submenus Memory Addressing

DF Watchdog Timer Interval	Selection	Auto [0xFF] ■ · 41 ms [0x0] · 166 ms [0x1] · 334 ms [0x2] · 669 ms [0x3] · 1.34 seconds [0x4] · 2.68 seconds [0x5] · 5.36 seconds [0x6]
Disable DF to external downstream IP SyncFloodPropagation	Selection	Sync flood disabled [0x1] · Sync flood enabled [0x0] · Auto [0xFF] ■
Sync Flood Propagation to DF Components	Selection	Sync flood disabled [0x1] · Sync flood enabled [0x0] · Auto [0xFF] ■
Freeze DF module queues on error	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■
CC6 memory region encryption	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■
System probe filter	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■ ■ KEEP Enabled (Auto) — Probe filter reduces coherency traffic; helps multi-core memory performance.
Disable DF sync flood propagation	Selection	Sync flood disabled [0x1] · Sync flood enabled [0x0] · Auto [0x3] ■
DF Cstates	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Disabled for lowest-latency / sustained all-core work (keeps Infinity Fabric out of low-power state).

Memory Addressing (0x7015)

Memory interleaving	Selection	Disabled [0x0] · Auto [0x7] ■ ■ KEEP Auto — Channel interleaving maximizes memory bandwidth; Disabled costs throughput on every workload.
Memory interleaving size	Selection	256 Bytes [0x0] · 512 Bytes [0x1] · 1 KB [0x2] · 2 KB [0x3] · Auto [0x7] ■ ■ KEEP Auto — Firmware picks the interleave granularity that suits the soldered configuration.
DRAM map inversion	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0x3] ■

Location of private memory regions	Selection	Distributed [0x0] · Consolidated [0x1] · Consolidated to 1st DRAM pair [0x2] · Auto [0xFF] ■
NUMA Nodes	Selection	NPS1 [0x1] ■ · NPS2 [0x2] · NPS4 [0x4] ■ KEEP NPS1 (single node) — Strix Halo presents one unified memory domain across both CCDs on a 256-bit LPDDR5X interface.

UMC Common Options (0x7003)

Submenus LPDDR Options

LPDDR Options (0x7019)

Submenus LPDDR Timing Configuration ■ LPDDR Controller Configuration ■ LPDDR Bus Configuration ■ LPDDR RAS ■ LPDDR Security ■ LPDDR Addressing Options ■ LPDDR Training Options ■ LPDDR Memory MBIST

LPDDR Timing Configuration (0x701A)

Submenus Decline ■ Accept

Information fields AMD out-of-specification operation warning — Decline / Accept gate; damage from out-of-spec use is not covered by warranty.

Accept (0x7023)

Submenus LPDDR SPD Timing ■ LPDDR Non-SPD Timing

Active Memory Timing Settings	Selection	Auto [0xFF] ■ · Enabled [0x1]
Maximum Memory Data Clock Speed	Selection	Auto [0xFFFF] ■ · 1600MT/s [0x640] · 3200MT/s [0xC80] · 4267MT/s [0x10AB] · 5500MT/s [0x157C] · 6400MT/s [0x1900] · 7500MT/S [0x1D4C] · 8000MT/S [0x1F40] ■ TUNE Auto — LPDDR5X-8000 is soldered and pre-tuned; leave the SoC default.

LPDDR SPD Timing (0x7024)

Trcpage Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcpage	Numeric	Min 0x0 · Max 0xFFF · Default 0x0 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcwl Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcwl	Numeric	Min 0x9 · Max 0x16 · Default 0x9 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcl Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcl	Numeric	Min 0x8 · Max 0x21 · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcdrd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcdrd	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcdwr Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcdwr	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trp Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trp	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

TrpPb Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrpPb	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc	Numeric	Min 0x3C · Max 0x3DE · Default 0x138 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrfcPb Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrfcPb	Numeric	Min 0x3C · Max 0x3DE · Default 0x138 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

LPDDR Non-SPD Timing (0x7025)

Tras Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tras	Numeric	Min 0x15 · Max 0x3A · Default 0x15 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trc	Numeric	Min 0x1D · Max 0x87 · Default 0x39 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrcPb Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrcPb	Numeric	Min 0x1D · Max 0x87 · Default 0x39 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdS Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdS	Numeric	Min 0x4 · Max 0xC · Default 0x4 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdL	Numeric	Min 0x4 · Max 0xC · Default 0x4 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tfaw Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tfaw	Numeric	Min 0x6 · Max 0x36 · Default 0x1A ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

TwtrS Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrS	Numeric	Min 0x2 · Max 0xE · Default 0x2 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrL	Numeric	Min 0x2 · Max 0xE · Default 0x2 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twr Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twr	Numeric	Min 0xA · Max 0x51 · Default 0x12 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trtp Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trtp	Numeric	Min 0x5 · Max 0xE · Default 0x5 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcke Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcke	Numeric	Min 0x1 · Max 0x1F · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdScL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdScL	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdDd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdDd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrScL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

TwrwrScL	Numeric	Min 0x1 · Max 0x3F · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrDd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrDd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twrrd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twrrd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trdwr Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trdwr	Numeric	Min 0x1 · Max 0x1F · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

LPDDR Controller Configuration (0x701B)

Submenus LPDDR Power Options

LPDDR Refresh Mode	Selection	Auto [0xFF] ■ · All Banks [0x1] · Per Bank [0x2] ■ KEEP Auto — Refresh policy is tuned for the soldered LPDDR5X; leave the default.
LPDDR RFM	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Auto — Refresh-management feature; leave to firmware.
WCK Always On	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
DVFSC Mode	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Auto — Dynamic memory voltage/frequency scaling; leave to firmware.
RRW Memory Test Control	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]

LPDDR Power Options (0x7026)

Power Down Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Disabled can trim memory latency slightly under load; Enabled (Auto) saves idle power. The default suits most workloads.
Phy Low Power Disable	Selection	Auto [0xFF] ■ · 0 [0x0] · 1 [0x1]

LPDDR Bus Configuration (0x701C)

CA drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28]
CK drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28]
DQ drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28]
DQS drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28]
WCK drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28]

Processor CA ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Processor CK ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Processor CS ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Processor DQ ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Processor DQS ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Processor WCK ODT impedance	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · Disable [0x0]
Dram CA ODT impedance	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3] · RZQ/4 [0x4] · RZQ/5 [0x5] · RZQ/6 [0x6]
Dram CS ODT impedance	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3]
Dram DQ ODT impedance	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3] · RZQ/4 [0x4] · RZQ/5 [0x5] · RZQ/6 [0x6]
Dram WCK ODT impedance	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3] · RZQ/4 [0x4] · RZQ/5 [0x5] · RZQ/6 [0x6]
Dram Non-Target ODT impedance	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3] · RZQ/4 [0x4] · RZQ/5 [0x5] · RZQ/6 [0x6]
Dram Pull-Down drive strengths	Selection	Auto [0xFF] ■ · Disable [0x0] · RZQ/1 [0x1] · RZQ/2 [0x2] · RZQ/3 [0x3] · RZQ/4 [0x4] · RZQ/5 [0x5] · RZQ/6 [0x6]

LPDDR RAS (0x701D)

DRAM Read Link ECC Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Link ECC adds reliability with a small bandwidth cost. Disabled for maximum memory throughput; Enabled for data integrity.
DRAM Write Link ECC Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Link ECC adds reliability with a small bandwidth cost. Disabled for maximum memory throughput; Enabled for data integrity.
Disable Memory Error Injection	Selection	False [0x0] · True [0x1] · Auto [0xFF] ■

LPDDR Security (0x701E)

TSME	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0] ■ TUNE Transparent SME (full-memory encryption) adds a small bandwidth/latency cost.
Data Scramble	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xFF] ■

LPDDR Addressing Options (0x701F)

Chip Select Interleaving	Selection	Disabled [0x0] · Auto [0xFF] ■ ■ KEEP Auto — Interleaving across chip selects aids bandwidth; leave it on.
Bank Swap	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Auto — Bank remapping for access spreading; leave to firmware.
BankGroup Swap	Selection	Disabled [0x0] · Enabled [0x2] · Auto [0xFF] ■ ■ KEEP Auto — Bank-group swap can improve access parallelism; leave the default.
Address Hash Bank	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ KEEP Enabled (Auto) — Address hashing reduces bank conflicts; beneficial for bandwidth.
Address Hash CS	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0] ■ KEEP Enabled (Auto) — Chip-select hashing spreads accesses; leave it on.

LPDDR Training Options (0x7020)

DFE Read Training	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0]
DFE Write Training	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0]

LPDDR Memory MBIST (0x7021)

Submenus LPDDR Data Eye

MBIST Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■
MBIST Test Mode	Selection	Interface Mode [0x0] · Data Eye Mode [0x1] · Both [0x2] · Auto [0xFF] ■
MBIST Aggressors	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■

LPDDR Data Eye (0x7029)

Pattern Select	Selection	Auto [0xFF] ■ · PRBS [0x0] · SSO [0x1] · Both [0x2]
Pattern Length Select	Selection	Auto [0xFF] ■ · Manual [0x1]
Pattern Length	Numeric	Min 0x3 · Max 0xC · Default 0x3 ■

Aggressor Channel	Selection	Auto [0xFF] ■ · Disabled [0x0] · 1 Aggressor Channel [0x1] · 3 Aggressor Channels [0x3] · 7 Aggressor Channels [0x7]
Read Voltage Sweep Step Size	Selection	Auto [0xFF] ■ · 1 [0x0] · 2 [0x1] · 4 [0x4]
Read Timing Sweep Step Size	Selection	Auto [0xFF] ■ · 1 [0x1] · 2 [0x2] · 4 [0x4]
Write Voltage Sweep Step Size	Selection	Auto [0xFF] ■ · 1 [0x0] · 2 [0x1] · 4 [0x4]
Write Timing Sweep Step Size	Selection	Auto [0xFF] ■ · 1 [0x1] · 2 [0x2] · 4 [0x4]

NBIO Common Options (0x7004)

Submenus GFX Configuration ■ Audio Configuration

IOMMU	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■ ■ TUNE Enabled (Auto) — Disabled measures ~6% higher iGPU memory read (234 vs 221 GB/s, community strix-halo-testing runs by lhl) but costs NPU, VFIO/GPU passthrough, DMA isolation and reliable suspend; Auto resolves to Enabled.
DMAr Support	Selection	Auto [0xF] ■ · Enabled [0x1] · Disabled [0x0]
DMA Protection	Selection	Auto [0xF] ■ · Enabled [0x1] · Disabled [0x0]
SCPC attribute control	Selection	0 [0x0] · 1 [0x1] · 2 [0x2] · 3 [0x3] · Customized [0xFF] ■
PCIe ARI Support	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■ ■ KEEP Enabled (Auto) — Alternative Routing-ID improves multi-function PCIe device support; leave it on.
PCIe loopback Mode	Selection	Auto [0xF] ■ · Disabled [0x0] · Enabled [0x1]
PSPP Policy	Selection	Auto [0xF] ■ · Disabled [0x0] · Balanced [0x2] ■ TUNE Balanced (Auto) for daily use; Disabled keeps PCIe links at full speed (lower latency, higher idle power).

GFX Configuration (0x702C)

iGPU Configuration	Selection	Auto [0xF] · iGPU Disabled [0x0] · UMA_SPECIFIED [0x1] ■ · UMA_AUTO [0x2] · UMA_GAME_OPTIMIZED [0x3] · UMA_WORKSTATION_OPTIMIZED [0x4] ■ KEEP UMA_SPECIFIED — Pairs with the UMA Frame buffer Size choice below.
iGPU Mem Bar Configuration	Selection	Auto [0xF] ■ · LegacyGfxBAR [0x1] · LargeBAR [0x2] · ResizableBAR [0x3] ■ KEEP Auto — Large iGPU BAR pairs with Above-4G/ReBAR for full-aperture access.
UMA Version	Selection	Legacy [0x0] · Non-Legacy [0x1] · Auto [0xF] ■
UMA Frame buffer Size	Selection	512M [0x200] · 4G [0x1000] · 8G [0x2000] · 16G [0x4000] · 32G [0x8000] · 64G [0x10000] · 96G [0x18000] ■ ■ CHANGE 512M [0x200] under Linux — RADV/Vulkan and ROCm reach memory through GTT, so a large carve-out only hides RAM from the OS (96G default leaves ~31 GiB of 128 GB visible).
GPU Host Translation Cache	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■ ■ KEEP Enabled (Auto) — Caches GPU address translations; helps iGPU memory-access latency.
TCON INSTANT ON LOGO	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Dedicated Graphics Memory Auto Level	Selection	UmaSizeMinimum [0x0] ■ · UmaSizeMedium [0x1] · UmaSizeHigh [0x2]
Remaining System Memory	Selection	RemainGB [0x0] ■

Audio Configuration (0x702D)

NB Azalia	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Audio IOs	Selection	Auto [0xFF] ■ · 21 options: HDA(3SDI) [0x1] … 2I2S + 1 INTR + PDM(4CH) [0x14]

FCH Common Options (0x7005)

Submenus I3C/I2C Configuration Options ■ USB Configuration Options ■ Ac Power Loss Options ■ Uart Configuration Options ■ USB4 Configuration Options ■ SPI Configuration Options

I3C/I2C Configuration Options (0x7037)

I3C/I2C 0 Enable	Selection	Both Disabled [0x3] · I3C Enabled [0x0] · I2C Enabled [0x1] · Auto [0xF] ■
I3C 0 Mode	Selection	I3C [0x0] · I2C [0x1] · Auto [0xF] ■
I3C/I2C 1 Enable	Selection	Both Disabled [0x3] · I3C Enabled [0x0] · I2C Enabled [0x1] · Auto [0xF] ■
I3C 1 Mode	Selection	I3C [0x0] · I2C [0x1] · Auto [0xF] ■
I3C/I2C 2 Enable	Selection	Both Disabled [0x3] · I3C Enabled [0x0] · I2C Enabled [0x1] · Auto [0xF] ■
I3C 2 Mode	Selection	I3C [0x0] · I2C [0x1] · Auto [0xF] ■
I3C/I2C 3 Enable	Selection	Both Disabled [0x3] · I3C Enabled [0x0] · I2C Enabled [0x1] · Auto [0xF] ■

I3C 3 Mode	Selection	I3C [0x0] · I2C [0x1] · Auto [0xF] ■

USB Configuration Options (0x7038)

Submenus USB0 2.0 port enable ■ USB1 2.0 port enable ■ USB2 2.0 port enable ■ USB3 2.0 port enable ■ USB0 3.1 port enable ■ USB1 3.1 port enable ■ USB2 3.1 port enable ■ USB3 3.1 port enable

USB0 controller enable	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xF] ■
USB1 controller enable	Selection	Enabled [0x1] · Disabled [0x0] · Auto [0xF] ■

USB0 2.0 port enable (0x703D)

USB0 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
USB0 2.0 Port 1	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
USB0 2.0 Port 2	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
USB0 2.0 Port 3	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
USB0 2.0 Port 4	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB1 2.0 port enable (0x703E)

USB1 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB2 2.0 port enable (0x703F)

USB2 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB3 2.0 port enable (0x7040)

USB3 2.0 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB0 3.1 port enable (0x7041)

USB0 3.1 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
USB0 3.1 Port 1	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB1 3.1 port enable (0x7042)

USB1 3.1 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB2 3.1 port enable (0x7043)

USB2 3.1 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB3 3.1 port enable (0x7044)

USB3 3.1 Port 0	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

Ac Power Loss Options (0x7039)

Ac Loss Control	Selection	Always Off [0x0] ■ · Always On [0x1] · Last State [0x3]

Uart Configuration Options (0x703A)

Uart 0 Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Uart 0 Legacy Options	Selection	Disabled [0x0] · 0x2E8 [0x1] · 0x2F8 [0x2] · 0x3E8 [0x3] · 0x3F8 [0x4] · Auto [0xF] ■
Uart 1 Enable (no HW FC)	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Uart 1 Legacy Options	Selection	Disabled [0x0] · 0x2E8 [0x1] · 0x2F8 [0x2] · 0x3E8 [0x3] · 0x3F8 [0x4] · Auto [0xF] ■
Uart 2 Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Uart 2 Legacy Options	Selection	Disabled [0x0] · 0x2E8 [0x1] · 0x2F8 [0x2] · 0x3E8 [0x3] · 0x3F8 [0x4] · Auto [0xF] ■
Uart 3 Enable (no HW FC)	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Uart 3 Legacy Options	Selection	Disabled [0x0] · 0x2E8 [0x1] · 0x2F8 [0x2] · 0x3E8 [0x3] · 0x3F8 [0x4] · Auto [0xF] ■
Uart 4 Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB4 Configuration Options (0x703B)

Submenus USB4 RT0 ■ USB4 RT1

USB4 pre-CM Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB4 RT0 (0x7045)

RT0 Router Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
RT0 PCIe Tunnel	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

USB4 RT1 (0x7046)

RT1 Router Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
RT1 PCIe Tunnel	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

SPI Configuration Options (0x703C)

HFP Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
HID Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

HID2 Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
HID Spi Read Mode	Selection	Quad IO 1-4-4 [0x5] ■ · Dual IO 1-2-2 [0x4] · Fast Read 1-1-1 [0x7]
HID Spi Speed	Selection	(1)33HMz [0x1] ■ · (2)22NMz [0x2] · (3)17.6Mhz [0x3] · (5)800Khz [0x5] · spi_spd6 [0x6] · spi_spd7 [0x7]
HID Spi Speed6 N	Numeric	Min 0x4 · Max 0x3F · Default 0x4 ■
HID Spi Speed7 N	Numeric	Min 0x4 · Max 0x3F · Default 0x32 ■
HID2 Spi Read Mode	Selection	Quad IO 1-4-4 [0x5] ■ · Dual IO 1-2-2 [0x4] · Fast Read 1-1-1 [0x7]
HID2 Spi Speed	Selection	(1)33HMz [0x1] ■ · (2)22NMz [0x2] · (3)17.6Mhz [0x3] · (5)800Khz [0x5] · spi_spd6 [0x6] · spi_spd7 [0x7]
HID2 Spi Speed6 N	Numeric	Min 0x4 · Max 0x3F · Default 0x4 ■
HID2 Spi Speed7 N	Numeric	Min 0x4 · Max 0x3F · Default 0x32 ■

SMU Common Options (0x7006)

System Configuration	Selection	120W [0x3] ■ · 12 options: 15W [0xE] … 140W [0x5] ■ TUNE Platform cTDP profile. Default 120W [0x3]; 140W [0x5] is the highest exposed and matches Beelink's validated 140 W ceiling. AMD's own cTDP range for this part is 45–120 W.
ECO Mode	Selection	Auto [0xF] ■ · Enabled [0x1] · Disabled [0x0] ■ TUNE Disabled for maximum performance. ECO caps APU power for efficiency/quiet at the cost of sustained clocks.
SPL Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to raise the Sustained Power Limit; a higher SPL sustains higher all-core/iGPU clocks within thermal limits.
Sustained Power Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Raise toward the GTR9 Pro ceiling (~140 W; value in mW) for best sustained throughput; bounded by cooling and skin-temperature limits.
PPT Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to lift Fast/Slow PPT limits for more boost headroom (cooling permitting).
Fast PPT Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Short-window PPT in mW; 0 means SMU default. Set only alongside Slow PPT and cTDP.
Slow PPT Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Averaged PPT in mW over the Slow PPT Time Constant; 0 means SMU default. Long loads are governed by the Sustained Power Limit.
Slow PPT Time Constant	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
STAPM Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to relax STAPM (skin-temperature-aware power) for more sustained performance; the default protects chassis temperature.
System Temperature Tracking	Selection	Auto [0xF] ■ · 1 [0x1] · 0 [0x0] ■ TUNE 0 disables skin-temperature tracking and removes a sustained-power cap (more performance, hotter chassis); leave Auto for safe thermals.
STAPM Boost Override	Selection	Auto [0xF] ■ · 1 [0x1] · 0 [0x0] ■ TUNE Set 1 to extend boost duration before STAPM clamps; raises chassis temperature. Auto follows the firmware policy.
STAPM Boost	Selection	Auto [0xF] ■ · 1 [0x1] · 0 [0x0] ■ TUNE 1 enables STAPM boost (a longer high-power window) once STAPM Boost Override is 1; bounded by chassis thermals.
Tskin Time Constant (STAPM)	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
Thermal Control	Selection	Manual [0x1] ■ · Auto [0x0] ■ TUNE Manual is the compiled default and exposes the TjMax row below; Auto returns the limit to firmware policy. Raise TjMax only with strong cooling.
TjMax	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x5A ■ ■ TUNE Leave the compiled default (0x5A). AMD rates Tjmax at 100 °C for the 395; lowering it reduces boost, and raising it past that rating is not advisable.
TDC Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to raise sustained current limits in step with PPT for all-core boost.

TDC_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Core-rail sustained current cap; raise it with PPT for sustained all-core clocks.
TDC_VDDCR_SOC	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
TDC_VDDCR_SR	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
EDC Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to raise peak current limits for burst boost.
EDC_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Core-rail peak current cap; raise it for burst boost headroom.
EDC_VDDCR_SOC	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
EDC_VDDCR_SR	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
PSI3 Control	Selection	Manual [0x1] · Auto [0x0] ■
PSI3_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
PROCHOT Control	Selection	Manual [0x1] · Auto [0x0] ■
PROCHOT Deassertion Ramp Time	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
STT Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Skin-temperature tracking master switch; Manual to raise the skin-temperature limits for more sustained power (hotter chassis).
STT_PCB_SENSOR_COUNT	Numeric	Min 0x0 · Max 0xF · Default 0x0 ■
STT_MIN_POWER_LIMIT	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M1	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M3	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M4	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M5	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_M6	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_C_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_C_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_ALPHA_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_ALPHA_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_SKIN_TEMPERATURE_LIMIT_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ ■ TUNE Raising the APU skin-temperature limit sustains boost longer at higher chassis temperature; the default protects surface temperature.
STT_SKIN_TEMPERATURE_LIMIT_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ ■ TUNE Raising the secondary skin-temperature limit sustains more power; the chassis runs hotter.
STT_ERROR_COEFF	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
STT_ERROR_RATE_COEFF	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
Fan Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual — Use a graduated aggressive curve. More airflow buys thermal headroom and higher sustained boost, the largest sustained-performance gain available on this chassis.
Force PWM Control	Selection	Force [0x1] · Unforce [0x0] ■ ■ TUNE Force pins a fixed PWM duty for maximum cooling during benchmarks or sustained load, at the cost of noise.
Force PWM	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ ■ TUNE Higher fixed PWM raises airflow and sustained clocks at the cost of noise.
Fan Table Control	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to define an aggressive temperature/PWM table for better sustained thermals.
Low Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.

Medium Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
High Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Critical Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
Low Pwm	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Medium Pwm	Numeric	Min 0x0 · Max 0x65 · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
High Pwm	Numeric	Min 0x0 · Max 0x66 · Default 0x0 ■ ■ TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temperature / higher PWM) to gain thermal headroom.
Temperature Hysteresis	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ ■ TUNE Deadband before the fan steps back down. Larger values reduce fan hunting; smaller values shed heat sooner.
Pwm Frequency	Selection	Auto [0xF] ■ · 1 [0x1] · 0 [0x0]
Fan polarity	Selection	Auto [0xF] ■ · 1 [0x1] · 0 [0x0]

SOC Miscellaneous Control (0x7007)

Submenus Pluton Options ■ Firmware Anti-rollback (FAR) ■ AIM-T Options ■ Intrusion Detection ■ SecureBio

Pluton Attribute Control	Selection	Disabled [0x0] · Enabled [0x1] ■
Pluton fTPM Attribute Control	Selection	Disabled [0x0] · Enabled [0x1] ■
Trusted Platform Module	Selection	Auto [0xFE] ■ · Disabled [0xFF] · Enable dTPM [0x0] · Enable ASP fTPM [0x1] · Enable Pluton fTPM [0x2]
Pluton Support	Selection	Auto [0xF] ■ · Disabled [0x0] · Enabled [0x1]
Pluton Security Processor	Selection	Auto [0xF] ■ · Disabled [0x0] · Enabled [0x1]
Microsoft Security Levels	Selection	Customized [0xFF] ■ · 18 options: dTPM Level 1 without Pluton Security Processor [0x1] … ASP fTPM Level 2 with Pluton Security Processor [0x34]
Secured-core Auto enablement	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]
DRTM Support	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]
SMM Isolation Support	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]
Pro Part attribute control	Selection	0 [0x0] ■ · 1 [0x1]
GFX workstation support	Selection	Auto [0xF] ■ · Enabled [0x1] · Disabled [0x0]
ABL Console Out Control	Selection	Auto [0x2] ■ · Enable [0x1] · Disable [0x0]
ABL Console Out Serial Port	Selection	Auto [0xFF] ■ · LPC UART [0x0] · FCH UART0 [0x1] · FCH UART1 [0x2] · FCH UART2 [0x3] · FCH UART3 [0x4]
ABL Console Out Boot Mode Select	Selection	ALL [0xFF] ■ · S3/S0i3 only [0x2] · Normal boot only [0x1]
ABL PMU message Control	Selection	Auto [0xFF] ■ · Maximal debug messages [0x4] · Detailed debug message [0x5] · Coarse debug message [0xA] · Stage completion [0xC8] · Assertion message [0xC9] · Firmware completion message only [0xFE]
PSP RPMC Switch	Selection	Auto [0xFF] ■ · Disabled [0x0] · Enabled [0x1]
ABL Training Mode	Selection	default [0x0] ■ · serial [0x1] · parallel [0x2]

Pluton Options (0x7056)

Pluton UART	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Pluton UART Port	Selection	UART 0 [0x0] · UART 1 [0x1] · UART 2 [0x2] · UART 3 [0x3] · UART 4 [0x4] · UART 5 [0x5] · UART 6 [0x6] · UART 7 [0x7] · Auto [0xF] ■
Pluton FIPS	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Pluton FIPS zeroization	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

Firmware Anti-rollback (FAR) (0x7057)

FAR enforcement state	Selection	Enabled [0x1] · Disabled [0x0] ■
SPL value in the CPU fuse	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■

SPL value in the SPL table	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
Initial SPL value for FAR	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
SPL update switch	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]

AIM-T Options (0x7058)

AIM-T Support	Selection	AIM-T Disabled [0x0] · AIM-T Enabled [0x1] · Auto [0xF] ■
MPM attribute control	Selection	0 [0x0] ■ · 1 [0x1] · 2 [0x2] · 3 [0x3] · 4 [0x4] · 5 [0x5] · 6 [0x6] · 7 [0x7]
KVM for Wired Manageability	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■
Wireless Manageability	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xF] ■

Intrusion Detection (0x7059)

Intrusion Detection Control	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
Intrusion Detection Enable	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0]
Log Intrusion Event	Selection	Enabled [0x1] · Disabled [0x0] ■
Clear TPM	Selection	Enabled [0x1] · Disabled [0x0] ■
Power Off System	Selection	Enabled [0x1] · Disabled [0x0] ■

SecureBio (0x705A)

SecureBio Support	Selection	Auto [0xF] ■ · Enable [0x1] · Disable [0x0]
SecureBio Camera Support	Selection	Disabled [0x0] ■ · XHCI Camera [0x1] · MIPI Camera [0x2] · Auto [0xF]

AMD PBS (Platform BIOS Settings)

14 forms · 202 settings

AMD PBS Option (0xB)

Submenus AMD Firmware Version ■ PCI Express Configurations ■ Power Saving Configurations ■ Graphics Configurations ■ Display Configurations ■ USB/Thunderbolt Configurations ■ USB TypeC Security Mode Configurations ■ Audio Configurations ■ I2C Configurations ■ Thermal Configurations ■ MP2 Configurations ■ EC/PD Configurations ■ Debug Configurations

Enable SLT check	Selection	Disabled [0x0] · Enabled [0x1] ■ · Force SLT check halt [0x2]
Above 4GB MMIO Limit	Selection	40bit (1TB) [0x28] ■ · 14 options: 35bit (32GB) [0x23] … 48bit (256TB) [0x30] ■ KEEP 40bit (1TB) — This sizes the above-4G MMIO aperture; it is not the Above-4G Decoding enable.
Wireless LAN Recovery	Selection	Disabled [0x0] ■ · Enabled [0x1] · Dummy reset [0x2]
Wireless LAN Country Code	Selection	Disabled [0x0] ■ · Country Code USA [0x1] · Country Code China [0x2]
Bluetooth PLDR	Selection	Disabled [0x0] ■ · Enabled [0x1] · Dummy reset [0x2]
Wireless Button	Selection	Disabled [0x0] ■ · HID Based [0x3]
Processor Aggregator Device	Selection	Disabled [0x0] ■ · Enabled [0x1]
Core Count Control	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ KEEP Disabled (all cores active) — Reducing core count lowers multi-thread performance; only down-core for special validation.
Core Count SW_SCI_GPE_ID	Numeric	Min 0x0 · Max 0x1F · Default 0x19 ■
APIC Software Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
Dynamic P3T limit	Selection	Disabled [0x0] · Enable for DC-only case (include fake DC) [0x1] ■ · Enabled [0x2] ■ TUNE Platform power-tracking limit. The compiled default keeps it dynamic on DC only; Enabled extends that to AC and Disabled pins P3T. Leave the default unless you set power limits by hand.
Dynamic LID	Selection	Disabled [0x0] ■ · Enabled [0x1]
HDMI 3.0G Tx SLEW	Selection	Disabled [0x0] ■ · Enabled [0x1]
HDMI 3.0G Tx Slew Control Value	Numeric	Min 0x0 · Max 0xFF · Default 0x14 ■
Control Method Button Support	Selection	Legacy Power Button (PNP0C0C) [0x0] ■ · Generic Button (ACPI0011) [0x1]

AMD Firmware Version (0xC)

Information fields AGESA Version, PSP BootLoader Version, PSP SecureOS Version, ABL Version, APCB Version, APOB Version, DF Rib Version, Ucode Patch Version, S3 Image ID, SMU FW Version, ZSC FW Version, MPIO FW Version, MP2 FW Version, XHCI FW Version, IMU FW Version, VBIOS FW Version, GOP Driver Version, EC FW Version, EDKII Version, PD0 FW Version, PD0 Custom, PD0 SKU, PD0 Mini Type, Re-timer, PD1 FW Version, PD1 Custom, PD1 SKU, PD1 Mini Type, AIM-T(MPM) Part1 FW Version, AIM-T(MPM) Part2 FW Version

PCI Express Configurations (0xD)

Pcie Dxio Timing ControlEnable	Selection	Auto [0xF] ■ · Enabled [0x1] · Disabled [0x0]
PCIE Link Receiver Detection Polling	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
PCIE Link L0 Polling	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
Enable power sequence of specific WWAN	Selection	Disabled [0x0] ■ · Power Sequence — Fibocom [0x1] · Power Sequence — Quectel RM520 [0x2] · Power Sequence — Quectel EM160 [0x3]
PCIE x4 DT Slot Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
M.2 SSD1 Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
WLAN Power Enable	Selection	Disabled [0x0] · x1 [0x1] ■ · x2 [0x2]
WWAN Power Enable	Selection	Disabled [0x0] · x1 [0x1] ■ · x2 [0x2]
SD Card Reader Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
Initial SD7 L1 Substates at OS runtime	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE PCIe L1-substate (L1.1/L1.2) power management for the SD7 card reader at OS runtime. Disabled (the default) keeps the lowest link exit latency; Enabled saves idle power.
GbE Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
M.2 SSD0 Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
NVMe RAID mode	Selection	Disabled [0x0] ■ · Enabled [0x1]
Save Restore DXIO Feature	Selection	Auto [0xF] ■ · Disabled [0x0] · Enabled [0x1]

Power Saving Configurations (0xE)

NVME D3Cold	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
M.2 NVME Tpvperl	Numeric	Min 0x32 · Max 0x3E8 · Default 0x32 ■
M.2 NVME Trst-cfg	Numeric	Min 0x64 · Max 0x3E8 · Default 0x64 ■
WLAN D3Cold	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
LOM D3Cold	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SD CardReader D3Cold	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
USB4 D3 Eanble	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Internal PCIe GPP 0 D3	Selection	Disabled [0x0] ■ · Enabled [0x4] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC GPU D3	Selection	Disabled [0x0] ■ · Enabled [0x4] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC HD Audio D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC USB3.1 D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC ACP D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC Azalia D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Internal PCIe GPP 2 D3	Selection	Disabled [0x0] ■ · Enabled [0x4] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC USB3.1 for USB4 D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC USB4 D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Internal USB4 PCIe Tunneling D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
SOC USB4 PCIe Endpoint D3	Selection	Disabled [0x0] · Enabled [0x4] ■ ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Keep Wlan Power In S3/S4 state	Selection	Disabled [0x0] ■ · Enable Wlan Power In S3 And S4 [0x3]
Unused GPP Clocks Off	Selection	Disabled [0x0] ■ · Enabled [0x1]
Clock PM: CLK_REQ0	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).

Clock PM: CLK_REQ1	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
Clock PM: CLK_REQ2	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
Clock PM: CLK_REQ3	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
Clock PM: CLK_REQ4	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
Clock PM: CLK_REQ5	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
Clock PM: CLK_REQ6	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ TUNE Clock power management (CLKREQ#). Disabled keeps the PCIe reference clock running for the lowest L1 exit latency (best performance, higher idle power).
PCIe x4 Slot D3 Cold	Selection	Disabled [0x0] ■ · Renesas XHCI [0x1] · Realtek SD Card Reader [0x2] ■ TUNE D3Cold gating for the device behind the x4 slot. Disabled keeps it out of D3Cold (lowest wake latency); the other options name which device may enter it.

USB/Thunderbolt Configurations (0x17)

PD USB4 Control Enable/Disable	Selection	Disabled [0x0] · Enabled [0x1] ■
USB4 Bus Reserved	Numeric	Min 0x0 · Max 0xFF · Default 0x60 ■
USB4 IO Reserved (KB)	Numeric	Min 0x0 · Max 0xFF · Default 0x10 ■
USB4 Non-Prefetch Memory Reserved	Numeric	Min 0x0 · Max 0xFFFF · Default 0x180 ■
USB4 non-Prefetch MMIO align (0 ~ 65534 MB)	Numeric	Min 0x0 · Max 0xFFFF · Default 0x40 ■
USB4 Prefetch Memory Reserved	Selection	128GB [0x20000] ■ · 17 options: 2MB [0x2] … 64GB [0x10000]
USB4 Prefetch MMIO align	Selection	32GB [0x8000] ■ · 16 options: 2MB [0x2] … 64GB [0x10000]
USB Camera Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
USB Fingerprint or GBE MUX	Selection	Fingerprint [0x0] · GBE [0x1] ■
USB Fingerprint Enable	Selection	Disabled [0x0] ■ · Enabled [0x1]
UCSI Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
UCSI tunnel location	Selection	UCSI tunnel at EC RAM [0x0] · UCSI tunnel at MMIO 0xFEEC2000 [0x1] ■
USBC Port Harware Disable Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Reconfig Rebalance Resources by OS	Selection	Disabled [0x0] ■ · Enabled [0x1]
PD Thunderbolt3 Alt Mode	Selection	Enabled [0x0] ■ · Disabled [0x1]
USB4 ACPI _DEP Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
USB4 LTR Snoop/NonSnoop Scale	Numeric	Min 0x0 · Max 0x7 · Step 0x1 · Default 0x2 ■
USB4 LTR Snoop/NonSnoop Value	Numeric	Min 0x1 · Max 0x3FF · Step 0x1 · Default 0xC8 ■
Modify _UPC/_PLD settings for USB3.2-A vertical port (Requires board rework)	Selection	Disabled [0x0] ■ · Enabled [0x1]

USB TypeC Security Mode Configurations (0x18)

USB Type-C Security	Selection	Auto [0xF] ■ · Mode 0 [0x0] · Mode 1 [0x1] · Mode 2 [0x2]

Graphics Configurations (0xF)

EVAL Slot Power Enable	Selection	Disabled [0x0] · Enabled [0x1] ■
EVAL CARD T-Diode Routing Select	Selection	APU SMBUS1 [0x0] · EC [0x1] ■
Special Display Features	Selection	Disabled [0x0] · HybridGraphics [0x4] ■

D3Cold Support	Selection	Disabled [0x0] ■ · Enabled [0x1] · Dummy D3Cold [0x2] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Discrete GPU Hotplug Mode	Selection	Basic Mode [0x0] · Enhanced Mode [0x1] ■ · Non-Hotplug Mode [0x2]
Discrete GPU D3Cold HPD Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
PME Turn Off Support	Selection	Disabled [0x0] · Enabled [0x1] ■
D3Cold Force Gen1	Selection	Disabled [0x0] ■ · Enabled [0x1]
SLOTPWR-PWREN Timing(ms)	Numeric	Min 0x0 · Max 0x32 · Step 0x1 · Default 0x1 ■
PWREN-RST Timing (ms)	Numeric	Min 0x0 · Max 0x96 · Step 0x1 · Default 0x50 ■
PERST-WAKEL23 Timing (ms)	Numeric	Min 0x1 · Max 0x1E · Default 0xA ■
DLACT-CFGACC Timing (ms)	Numeric	Min 0x1 · Max 0x64 · Default 0x64 ■
NVIDIA DGPU Power Enable	Selection	Disabled [0x0] ■ · Enabled [0x1]
Discrete GPU _DSM Function A	Selection	Disabled [0x0] ■ · Enabled [0x1]
Discrete GPU _DSM Function B	Selection	Disabled [0x0] ■ · Enabled [0x1]
Non-Eval Discrete GPU Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Discrete GPU HPD Circuitry	Selection	OR Circuitry [0x0] ■ · Pulse Circuitry [0x1]
Discrete GPU's USB Port	Selection	Keep Default Setting [0x0] ■ · Disabled [0x1]
Discrete GPU's SSID/SVID	Selection	Keep Default Setting [0x0] · Program by Vendor [0x1] ■
Discrete GPU's VGA SSID/SVID	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
Discrete GPU's AUDIO SSID/SVID	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■
Discrete GPU BOMACO Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
MACO-PWR Timing (us)	Numeric	Min 0xA · Max 0xFF · Step 0x1 · Default 0xA ■
RST-MACO Timing (ms)	Numeric	Min 0x0 · Max 0xA · Step 0x1 · Default 0x0 ■
ATCS Function 9 Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
AC Maximum Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ ■ TUNE AC power-profile cap for this OS power-slider tier (0–100, 0 = firmware default). Leave 0; only the AC tiers are exercised on this mains-powered mini-PC, and a lower value only caps that tier.
AC Better Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ ■ TUNE AC power-profile cap for this OS power-slider tier (0–100, 0 = firmware default). Leave 0; only the AC tiers are exercised on this mains-powered mini-PC, and a lower value only caps that tier.
AC Better Battery Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ ■ TUNE AC power-profile cap for this OS power-slider tier (0–100, 0 = firmware default). Leave 0; only the AC tiers are exercised on this mains-powered mini-PC, and a lower value only caps that tier.
DC Maximum Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■
DC Better Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■
DC Better Battery Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■
DC Battery Saver Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■
BLINK LED	Selection	Disabled [0x0] · Enabled [0x1] ■ · GPIO 11 Output Low [0x2] · GPIO 11 Output High [0x3]
ATIF Notify Command Code	Selection	Notify VGA 0x81 [0x0] ■ · Notify VGA 0xD0 [0xD0] · Notify VGA 0xD1 [0xD1] · Notify VGA 0xD2 [0xD2] · Notify VGA 0xD3 [0xD3] · Notify VGA 0xD4 [0xD4] · Notify VGA 0xD5 [0xD5] · Notify VGA 0xD6 [0xD6] · Notify VGA 0xD7 [0xD7] · Notify VGA 0xD8 [0xD8] · Notify VGA 0xD9 [0xD9]
ATIF Function 21 Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
External Graphics Port	Selection	Disabled [0x0] ■ · Enabled [0x1]
Hide XConnect GUI	Selection	Disabled [0x0] ■ · Enabled [0x1]
Run Time PM and D3	Selection	Disabled [0x0] ■ · Enabled [0x1] ■ TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal gain); Enabled saves idle power.
Support ATIF ATPX	Selection	Disabled [0x0] ■ · Enabled [0x1]
ATIF Function 22 Support	Selection	Disabled [0x0] ■ · Undefined [0x1] · Integrated Graphics [0x2] · Discrete Graphics [0x3]
GPU Package Power Limit Value	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Within the shared APU budget, a higher GPU power slice favors graphics-bound work.
ATIF Function 23 Support	Selection	Disabled [0x0] ■ · Enabled [0x1]

Vari-Bright Maximum Performance	Numeric	Min 0x0 · Max 0x4 · Default 0x0 ■
Vari-Bright Better Performance	Numeric	Min 0x0 · Max 0x4 · Default 0x1 ■
Vari-Bright Better Battery	Numeric	Min 0x0 · Max 0x4 · Default 0x3 ■
Vari-Bright Battery Saver	Numeric	Min 0x0 · Max 0x4 · Default 0x3 ■
ATIF Function 24 Support	Selection	Disabled [0x0] · Enabled [0x1] ■
ISP Device number	Numeric	Min 0x0 · Max 0x64 · Step 0x1 · Default 0x0 ■
Primary Video Adaptor	Selection	Int Graphics (IGD) [0x1] ■ · Ext Graphics (PEG) [0x2]
Smart Mux Acpi Control	Selection	Disabled [0x0] ■ · Enabled without _DEP [0x1] · Enabled [0x3]
Display Panel Multiplexer	Numeric	Min 0x0 · Max 0x7 · Default 0x0 ■
Smart Mux _HID Selection	Selection	SMUX1234 [0x0] ■ · MSFT0007 [0x1]
Smart Mux MDM Support Level	Selection	No Support [0x0] ■ · Development Support [0x1] · Experimental Support [0x2] · Full Support [0x3]
Smart Mux First Connected GPU	Selection	Integrated Graphics [0x0] ■ · Discrete Graphics [0x1]
Smart Mux ACPI Method Location	Selection	Under Mux [0x1] · Under dGPU [0x2] ■
Verify PEI GOP	Selection	Disabled [0x0] ■ · Enabled [0x1]
Ultra mode support	Selection	Disabled [0x0] ■ · Enabled [0x1]

Display Configurations (0x10)

DP0	Selection	Default [0x0] ■ · EDP display [0x1] · DP display [0x2] · HDMI display [0x3]
Adjust DP Caps	Selection	Soc Default [0x0] ■ · Cap override [0x1]
DP0 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■
DP1	Selection	Default [0x0] ■ · EDP display [0x1] · DP display [0x2] · HDMI display [0x3]
DP1 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■
DP2	Selection	Default [0x0] ■ · EDP display [0x1] · HDMI display [0x3] · DP with TypeC display [0x4] · DP without TypeC display [0x5]
DP2 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■
DP3	Selection	Default [0x0] ■ · EDP display [0x1] · HDMI display [0x3] · DP with TypeC display [0x4] · DP without TypeC display [0x5]
DP3 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■
DP4	Selection	Default [0x0] ■ · EDP display [0x1] · HDMI display [0x3] · DP with TypeC display [0x4] · DP without TypeC display [0x5]
DP4 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■
EDP priority	Selection	EDP0 [0x0] ■ · EDP1 [0x1]

Audio Configurations (0x11)

Wake On Voice	Selection	Disabled [0x0] ■ · Enabled [0x1]
ACP Power Gating	Selection	Disabled [0x0] · Enabled [0x1] ■ ■ KEEP Enabled — Power-gates the audio co-processor when idle; frees power budget with no downside.
Verb Table Select	Selection	ALC245 Crb Default: SVID/SSID: 1022/D670 [0x0] ■ · ALC245_05112020 : SVID/SSID: 10EC/1208 [0x1] · ALC256_06082020 : SVID/SSID: 10EC/1208 [0x2] · ALC623 : SVID/SSID: 10EC/1208 [0x3] · SN614X : SVID/SSID: 1022/D595 [0x4] · CX11970EVK : SVID/SSID: 14F1/0101 [0x5]
USB Sideband Audio	Selection	Disabled [0x0] ■ · Enabled [0x1]

I2C Configurations (0x12)

Touch Panel Support	Selection	Under I2C 0 Bus [0x0] · Under I2C 1 Bus [0x1] · Under I2C 2 Bus [0x2] · Under I2C 3 Bus [0x3] · ELAN TSP Under I2C 0 Bus [0x10] · ELAN TSP Under I2C 1 Bus [0x11] · ELAN TSP Under I2C 2 Bus [0x12] · ELAN TSP Under I2C 3 Bus [0x13] · Under USB 6 Bus [0x4] · Disabled [0x5] ■
Touch Pad Support	Selection	Under I2C 0 Bus [0x0] · Under I2C 1 Bus [0x1] · Under I2C 2 Bus [0x2] · Under I2C 3 Bus [0x3] · Disabled [0x4] ■
Touchpad Slave address	Numeric	Min 0x0 · Max 0xFF · Default 0x2C ■
Touchpad HID Descriptor address	Numeric	Min 0x0 · Max 0xFF · Default 0x20 ■
Nfc Support	Selection	Under I2C 0 Bus [0x0] · Under I2C 1 Bus [0x1] · Under I2C 2 Bus [0x2] · Under I2C 3 Bus [0x3] · Disabled [0x4] ■
MITT/WITT Selection	Selection	MITT Only [0x0] · WITT Only [0x1] · Both disable [0x2] ■
I2C MUX	Selection	Disabled [0x0] · Enabled [0x1] ■
SMBUS0 Buffer	Selection	Disabled [0x0] · Enabled [0x1] ■
SMBUS1 Buffer	Selection	Disabled [0x0] · Enabled [0x1] ■

Thermal Configurations (0x13)

APU PROCHOT# setting	Selection	Disable APU_PROCHOT# [0x0] · Enable APU_PROCHOT# in pure-DC case [0x1] · Enable APU_PROCHOT# in pure-AC case [0x2] · Enable APU_PROCHOT# in either pure-DC or pure-AC case (not AC+DC) [0x3] ■
AMD DPTC interface	Selection	Disabled [0x0] ■ · Enabled [0x1]
STT sensor reporting	Selection	Disabled [0x0] ■ · Report onboard sensors [0x1] · Report onboard + eval card sensors [0x2]

MP2 Configurations (0x14)

Power Sensors Routing Select	Selection	WALLE lite PDT [0x0] ■ · WALLE lite PM log [0x1]
MP2 FW Selection	Selection	MP2_SFH [0x0] ■ · MP2_WalleLite [0x1]
Turn off Xtal (S3/S5)	Selection	Disabled [0x0] · Enabled [0x1] ■
Sensor Fusion User Mode Driver	Selection	Disabled [0x0] · Enabled [0x1] ■
Reserved Memory For MP2	Selection	Disabled [0xFF] · 128 KB [0x0] ■ · 256 KB [0x1] · 512 KB [0x2] · 1 MB [0x3] · 2 MB [0x4] · 4 MB [0x5] · 8 MB [0x6] · 16 MB [0x7] · 32 MB [0x8] · 64 MB [0x9]
MP2 Privilege Mode	Selection	Disabled [0x0] ■ · Enabled [0x1]
Accelerator Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
Magnet Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
SRA Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
Light Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
Proximity Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
Wake On Human Presence detection WA	Selection	Disabled [0x0] ■ · Enabled [0x1]
Gyroscope Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]
HPD-Lite Sensor	Selection	Disabled [0x0] ■ · Enabled [0x1]

EC/PD Configurations (0x15)

Charger mode BYPASS	Selection	Disabled [0x0] ■ · Enabled [0x1]
KBC Support	Selection	Disabled [0x0] · Enabled [0x1] ■
AcDcSwitch	Selection	Disabled [0x0] ■ · Enabled [0x1]
AmdDcTimer	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
AmdAcTimer	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
Fake DC Level	Numeric	Min 0x0 · Max 0x64 · Default 0x5E ■
VDD adjust for BoardDesign	Selection	Disabled [0x0] ■ · Enabled [0x1]
VDDIO_MEM_S3 voltage	Selection	voltage (1.05V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDD MISC S5 voltage	Selection	voltage (0.75V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDD_MISC voltage	Selection	voltage (0.75V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDD11 voltage	Selection	voltage (1.1V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
1V8 ALW voltage	Selection	voltage(1.8V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDD2L voltage	Selection	voltage(0.9V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDDQ voltage	Selection	voltage(0.5V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
VDD MEM voltage	Selection	voltage(0.78V) [0x0] ■ · 21 options: voltage (+10mv) [0x1] … voltage (-100mv) [0x14]
POST LED on enable	Selection	Disabled [0x0] · Enabled [0x1] ■
Smart Mux Support	Selection	Disabled [0x0] ■ · Hybrid Graphics Mode [0x1] · Discrete Mode [0x2] · Smart Mux 1.5 [0x3] · Smart Mux 2.0 [0x4]
EC UDC Log Support	Selection	Disabled [0x0] · Enabled [0x1] ■
Retiemr I2C0/SMBUS1 Enable	Selection	Disabled [0x0] ■ · Enabled [0x1]
Retimer FMP driver support	Selection	Disabled [0x0] ■ · I2c Retimer FMP driver without force power [0x1] · I2c Retimer FMP driver with force power [0x2] · SBU Retimer FMP driver without force power [0x3] · SBU Retimer FMP driver with force power [0x4]

Debug Configurations (0x16)

Serial Debug Message Under OS	Selection	Disabled [0x0] ■ · Enabled [0x1]
Debug Print In ASL	Selection	Disabled [0x0] ■ · Enabled [0x1]
ISP Camera Module List	Selection	Disabled [0x0] ■ · 14 options: Module 1 [0x1] … Module 13 [0xD]
UEFI ISP I2c support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Debug Data Preserve	Selection	Disabled [0x0] ■ · Enabled [0x1]

Time for Blank Screen (10ms)	Numeric	Min 0x0 · Max 0x3E8 · Step 0x1 · Default 0xC8 ■

AMD Overclocking (AOD)

28 forms · 153 settings

AMD Overclocking Option (0x98)

Submenus Accept ■ Decline

AMD Overclocking (0x100)

Submenus Manual CPU Overclocking ■ Manual iGPU Overclocking ■ DDR and Infinity Fabric Frequency/Timings ■ Precision Boost Overdrive ■ VDDP Voltage Control ■ SoC/Uncore OC Mode ■ SoC Voltage ■ LN2 Mode ■ VDD Misc ■ LCLK Frequency Control ■ Onboard Voltage Control

Manual CPU Overclocking (0x101)

Submenus CPU Core Count Control

CPU Frequency	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x5 · Default 0x0 ■ ■ TUNE Auto (0) — Use PBO + Curve Optimizer rather than a fixed all-core frequency, which disables opportunistic boost.
CPU Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■ ■ TUNE Auto (0) — Manual Vcore disables adaptive boost; undervolt via Curve Optimizer instead.

CPU Core Count Control (0x10C)

SMT Control	Selection	Auto [0x1] ■ · Disable [0x0] ■ KEEP Auto — 32 threads. Disable only for a latency-sensitive workload that scales worse with SMT.
Down Core Mode	Selection	Enable [0x1] · Disable [0x0] ■ ■ KEEP Disable — Down-coring reduces multi-thread performance; enable it only for special validation.

Manual iGPU Overclocking (0x102)

GFX Clock Frequency	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ ■ TUNE Auto (0) — Fixing the iGPU clock removes dynamic boost; prefer GPU Boost Clock Override.
GFX Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■ ■ TUNE Auto (0) — Manual iGPU voltage is expert-only; Strix Halo has no confirmed iGPU undervolt.

DDR and Infinity Fabric Frequency/Timings (0x103)

Submenus DDR Options ■ Infinity Fabric Frequency and Dividers

DDR Options (0x110)

Submenus DDR Timing Configuration ■ DDR Bus Configuration ■ DDR Controller Configuration

DDR Timing Configuration (0x112)

Submenus DDR SPD Timing ■ DDR Non-SPD Timing

Active Memory Timing Settings	Selection	Auto [0xFF] ■ · Enabled [0x1]
Memory Target Speed	Selection	Auto [0xFFFF] ■ · 48 options: 2000 MTs [0x7D0] … 12000 MTs [0x2EE0] ■ TUNE Auto — Keep the soldered LPDDR5X at its rated 8000 MT/s; it is not user-replaceable and already runs at spec.

DDR SPD Timing (0x11B)

Tcl Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tcl	Numeric	Min 0x16 · Max 0x3E · Step 0x2 · Default 0x16 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trcd	Numeric	Min 0x8 · Max 0x3E · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trp Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

Trp	Numeric	Min 0x8 · Max 0x3E · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tras Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tras	Numeric	Min 0x1E · Max 0x7E · Default 0x1E ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trc	Numeric	Min 0x20 · Max 0xFF · Default 0x20 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twr Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twr	Numeric	Min 0x30 · Max 0x60 · Default 0x30 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc1 Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc1	Numeric	Min 0x32 · Max 0xFFF · Default 0x32 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc2 Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trfc2	Numeric	Min 0x32 · Max 0xFFF · Default 0x32 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrfcSb Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrfcSb	Numeric	Min 0x32 · Max 0x7FF · Default 0x32 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trtp Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trtp	Numeric	Min 0x5 · Max 0x1F · Default 0x9 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdL	Numeric	Min 0x4 · Max 0x20 · Default 0x4 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdS Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrrdS	Numeric	Min 0x4 · Max 0x14 · Default 0x4 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

Tfaw Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Tfaw	Numeric	Min 0x14 · Max 0x50 · Default 0x1A ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrL	Numeric	Min 0x8 · Max 0x30 · Default 0x8 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrS Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwtrS	Numeric	Min 0x2 · Max 0x10 · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

DDR Non-SPD Timing (0x11C)

TrdrdScL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdScL	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdSd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdDd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TrdrdDd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrScL Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrScL	Numeric	Min 0x1 · Max 0x3F · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSc Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

TwrwrSd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrSd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrDd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
TwrwrDd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twrrd Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Twrrd	Numeric	Min 0x1 · Max 0xF · Default 0x5 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trdwr Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.
Trdwr	Numeric	Min 0x1 · Max 0x1F · Default 0x5 ■ ■ TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom.

DDR Bus Configuration (0x113)

Processor CA drive strengths	Selection	Auto [0xFF] ■ · 120.0 Ohm [0x78] · 60.0 Ohm [0x3C] · 40.0 Ohm [0x28] · 30.0 Ohm [0x1E]
Processor DQ drive strengths	Selection	Auto [0xFF] ■ · High Impedance [0x0] · 240 ohm [0x2] · 120 ohm [0x4] · 80 ohm [0x6] · 60 ohm [0xC] · 48 ohm [0xE] · 40 ohm [0x1C] · 34.3 ohm [0x1E]
Processor ODT impedance	Selection	Auto [0xFF] ■ · 21 options: High Impedance [0x0] … 25.3 ohm [0x3F]
Dram DQ drive strengths	Selection	Auto [0xFF] ■ · 48 ohm [0x2] · 40 ohm [0x1] · 34 ohm [0x0]
Dram ODT impedance RTT_NOM_WR	Selection	Auto [0xFF] ■ · RTT_OFF [0x0] · RZQ (240) [0x1] · RZQ/2 (120) [0x2] · RZQ/3 (80) [0x3] · RZQ/4 (60) [0x4] · RZQ/5 (48) [0x5] · RZQ/6 (40) [0x6] · RZQ/7 (34) [0x7]
Dram ODT impedance RTT_NOM_RD	Selection	Auto [0xFF] ■ · RTT_OFF [0x0] · RZQ (240) [0x1] · RZQ/2 (120) [0x2] · RZQ/3 (80) [0x3] · RZQ/4 (60) [0x4] · RZQ/5 (48) [0x5] · RZQ/6 (40) [0x6] · RZQ/7 (34) [0x7]
Dram ODT impedance RTT_WR	Selection	Auto [0xFF] ■ · RTT_OFF [0x0] · RZQ (240) [0x1] · RZQ/2 (120) [0x2] · RZQ/3 (80) [0x3] · RZQ/4 (60) [0x4] · RZQ/5 (48) [0x5] · RZQ/6 (40) [0x6] · RZQ/7 (34) [0x7]
Dram ODT impedance RTT_PARK	Selection	Auto [0xFF] ■ · RTT_OFF [0x0] · RZQ (240) [0x1] · RZQ/2 (120) [0x2] · RZQ/3 (80) [0x3] · RZQ/4 (60) [0x4] · RZQ/5 (48) [0x5] · RZQ/6 (40) [0x6] · RZQ/7 (34) [0x7]
Dram ODT impedance DQS_RTT_PARK	Selection	Auto [0xFF] ■ · RTT_OFF [0x0] · RZQ (240) [0x1] · RZQ/2 (120) [0x2] · RZQ/3 (80) [0x3] · RZQ/4 (60) [0x4] · RZQ/5 (48) [0x5] · RZQ/6 (40) [0x6] · RZQ/7 (34) [0x7]

DDR Controller Configuration (0x114)

Submenus DDR Power Options

DDR Power Options (0x11D)

Power Down Enable	Selection	Disabled [0x0] · Enabled [0x1] · Auto [0xFF] ■ ■ TUNE Disabled can trim memory latency slightly under load; Enabled (Auto) saves idle power. The default suits most workloads.

Infinity Fabric Frequency and Dividers (0x111)

Infinity Fabric Frequency and Dividers	Selection	Auto [0xFFFF] ■ · 57 options: 333 MHz [0x14D] … 3000 MHz [0xBB8] ■ TUNE Auto — On Strix Halo the fabric/UCLK ratio is pre-balanced to the soldered LPDDR5X; manual changes rarely help and can destabilize.
UCLK DIV1 MODE	Selection	Auto [0xFF] ■ · UCLK=MEMCLK [0x1] · UCLK=MEMCLK/2 [0x0] ■ TUNE Auto (UCLK=MEMCLK, i.e., 1:1, where firmware allows it) — 1:1 minimizes memory latency; leave the firmware default for soldered RAM.

Precision Boost Overdrive (0x104)

Submenus GFX Curve Optimizer ■ Curve Optimizer ■ Curve Shaper

Precision Boost Overdrive	Selection	Auto [0xFF] ■ · Disable [0x0] · Enabled [0x1] · Advanced [0x2] ■ CHANGE Advanced (or Enabled) — Raises PPT/TDC/EDC ceilings so boost sustains longer. PBO is supported on the non-PRO 395 (AMD spec sheet); the PRO SKU has it fused off.
PBO Limits	Selection	Auto [0xFF] ■ · Disable [0x0] · Motherboard [0x1] · Manual [0x2] ■ CHANGE Motherboard — Lets the chip use the full board budget. Manual for explicit PPT/TDC/EDC.
PPT Limit [mW]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE AMD rates cTDP 45–120 W; Beelink validates 140 W sustained on this chassis. Value is in mW, so 140 W = 140000.
TDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Increase alongside PPT for sustained-load current headroom; keep within VRM/thermal capability.
EDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Increase for peak/burst current headroom; raising EDC mainly helps short bursty boost.
CCX (or second rail) TDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Secondary-rail TDC; raise it in step with the main TDC limit when pushing all-core.
CCX (or second rail) EDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ ■ TUNE Secondary-rail EDC; raise it in step with the main EDC limit for burst headroom.
Precision Boost Overdrive Scalar Ctrl	Selection	Auto [0xFF] ■ · Manual [0x1] ■ TUNE Manual only if raising the scalar; Auto keeps stock reliability limits.
Precision Boost Overdrive Scalar	Selection	1X [0x64] ■ · 2X [0xC8] · 3X [0x12C] · 4X [0x190] · 5X [0x1F4] · 6X [0x258] · 7X [0x2BC] · 8X [0x320] · 9X [0x384] · 10X [0x3E8] ■ TUNE 1X–2X for daily use; up to 10X sustains higher voltage/boost longer at the cost of heat and longevity. 2X–4X is a reasonable gaming target.
CPU Boost Clock Override	Selection	Disabled [0x0] ■ · Enabled (Positive) [0x1] · Enabled (Negative) [0x2] ■ TUNE Enabled (Positive) to allow up to +200 MHz above rated boost when thermals permit.
Max CPU Boost Clock Override(+)	Numeric	Min 0x19 · Max 0xC8 · Step 0x19 · Default 0x19 ■ ■ TUNE Add up to +200 MHz; effective only with thermal/power headroom (PBO + cooling).
Max CPU Boost Clock Override(-)	Numeric	Min 0x19 · Max 0x3E8 · Step 0x19 · Default 0x19 ■
GPU Boost Clock Override	Selection	Enabled [0x1] · Disabled [0x0] ■ ■ TUNE Enabled — Lifts the iGPU boost ceiling for graphics-bound gains, thermals permitting.
Max GPU Boost Clock Override	Selection	0 MHz [0x0] ■ · 25 MHz [0x19] · 50 MHz [0x32] · 75 MHz [0x4B] · 100 MHz [0x64] · 125 MHz [0x7D] · 150 MHz [0x96] · 175 MHz [0xAF] · 200 MHz [0xC8] ■ TUNE Add a modest iGPU boost offset and test; the 8060S is bandwidth/thermal-limited, so gains taper.
Platform Thermal Throttle Ctrl	Selection	Manual [0x1] · Auto [0x0] ■ ■ TUNE Manual to raise the throttle limit for more sustained clocks (only with strong cooling).
Platform Thermal Throttle Limit	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ ■ TUNE Raising the skin/platform throttle point sustains boost longer but increases temperatures and fan noise.

GFX Curve Optimizer (0x11E)

GFX Curve Optimizer	Selection	Disable [0x0] ■ · GFX Curve Optimizer [0x1] ■ TUNE GFX Curve Optimizer [0x1] arms the iGPU V/F offset below; unconfirmed on Strix Halo.
GFX Curve Optimizer Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative requests an iGPU undervolt; unconfirmed on Strix Halo, so measure the effect.
GFX Curve Optimizer Magnitude	Numeric	Min 0x0 · Max 0x32 · Default 0x0 ■ ■ TUNE Start small (10 with the Negative sign) and validate; iGPU stability is workload-sensitive.

Curve Optimizer (0x11F)

Curve Optimizer	Selection	Disable [0x0] ■ · All Cores [0x1] · Per Core [0x2] ■ CHANGE All Cores — Pair with the Negative sign below. Curve Optimizer is supported on the non-PRO 395 and is the highest-value PBO sub-setting.
All Core Curve Optimizer Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ CHANGE Negative — Undervolts the V/F curve for higher sustained boost and lower temperatures.
All Core Curve Optimizer Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Start at 10 with the Negative sign and step up while stress-testing; max is 0x1E (30). Validate under load, not idle.

Curve Shaper (0x126)

Min Frequency — Low Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Min Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Min Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Min Frequency — Med Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Min Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Min Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Min Frequency — High Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Min Frequency — High Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Min Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Low Frequency — Low Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Low Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Low Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Low Frequency — Med Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.

Low Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Low Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Low Frequency — High Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Low Frequency — High Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Low Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Med Frequency — Low Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Med Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Med Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Med Frequency — Med Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Med Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Med Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Med Frequency — High Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Med Frequency — High Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Med Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
High Frequency — Low Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
High Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.

High Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
High Frequency — Med Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
High Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
High Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
High Frequency — High Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
High Frequency — High Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
High Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Max Frequency — Low Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Max Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Max Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Max Frequency — Med Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Max Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Max Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.
Max Frequency — High Temperature	Selection	Auto [0xFF] ■ · Enable [0x1] · Disable [0x0] ■ TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region.
Max Frequency — High Temperature Sign	Selection	Positive [0x0] ■ · Negative [0x1] ■ TUNE Negative undervolts this frequency/temperature band.
Max Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ ■ TUNE Offset magnitude for this band (max 30). Start at 3–5 per band and validate.

VDDP Voltage Control (0x105)

VDDP Voltage Control	Selection	Auto [0xFF] ■ · Manual [0x1]

VDDP Voltage Adjust	Numeric	Min 0x0 · Max 0x7D0 · Default 0x0 ■

SoC/Uncore OC Mode (0x106)

SoC/Uncore OC Mode	Selection	Auto [0xFF] ■ · Enabled [0x1] · Disabled [0x0] ■ TUNE Auto — Forcing max SoC/uncore raises power and heat for little gain on this APU; leave the default.

SoC Voltage (0x107)

SoC Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■

LN2 Mode (0x108)

LN2 Mode	Selection	Auto [0xFF] ■ · Disabled [0x0] · Enabled [0x1]

VDD Misc (0x109)

VDD Misc Control	Selection	Auto [0x0] ■ · Manual [0x1]
VDD Misc Voltage	Numeric	Min 0x1F4 · Max 0x15E0 · Default 0x320 ■

LCLK Frequency Control (0x10A)

LCLK Frequency Control	Selection	Auto [0x0] ■ · Manual [0x1] ■ TUNE Auto — LCLK is the NBIO/PCIe fabric clock. Manual max can lower PCIe/IO latency but raises power; the default is balanced.
Minimun Frequency	Numeric	Min 0x96 · Max 0x9C4 · Default 0x96 ■ ■ TUNE LCLK floor. Raising the minimum keeps the NBIO/PCIe fabric clock high for lower IO latency at the cost of idle power; the default is balanced.
Maximum Frequency	Numeric	Min 0x96 · Max 0x9C4 · Default 0x9C4 ■ ■ KEEP LCLK ceiling; the compiled default is already the maximum (0x9C4), and LCLK Frequency Control Auto lets the IO fabric reach it under load.

Onboard Voltage Control (0x10B)

Submenus VDDIO Voltage Control ■ VPP Voltage Control ■ VDD_MEM Voltage Control

VDDIO Voltage Control (0x123)

Information fields Current DIMM VDD Voltage, Current DIMM VDDQ Voltage, Current APU VDDIO Voltage

VDDIO Ctrl	Selection	Auto [0x0] ■ · Manual [0x1] · Separate [0x2]
DIMM VDD Adjust	Numeric	Min 0x320 · Max 0x596 · Step 0xA · Default 0x44C ■
DIMM VDDQ Adjust	Numeric	Min 0x320 · Max 0x596 · Default 0x44C ■

VPP Voltage Control (0x124)

Information fields Current VPP Voltage

VPP Ctrl	Selection	Auto [0x0] ■ · Manual [0x1]
VPP Adjust	Numeric	Min 0x5DC · Max 0x852 · Step 0xA · Default 0x708 ■

VDD_MEM Voltage Control (0x125)

Information fields Current VDD_MEM Voltage

VDD_MEM Control	Selection	Auto [0x0] ■ · Manual [0x1]

AMD PMF (Platform Management Framework)

28 forms · 139 settings

AMD PMF Feature (0x64)

Submenus APMF Multiple Binaries Settings ■ APTS Control Method Settings ■ ATST Function ■ APMF Function 1 Settings ■ APMF Function 4 Settings ■ APMF Function 7 Settings ■ APMF Function 9 Settings ■ APMF Function 11 Settings ■ APMF Function 12 Settings ■ APMF Function 14 Settings ■ APMF Function 16 Settings

PMF Device Support	Selection	Disabled [0x0] ■ · Enabled [0x1]

APMF Multiple Binaries Settings (0x6D)

APMF Multiple Binaries Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
PMF Policy Binaries Index	Selection	PMF Binary 0 [0x0] ■ · PMF Binary 1 [0x1] · PMF Binary 2 [0x2] · PMF Binary 3 [0x3] · PMF Binary 4 [0x4] · PMF Binary 5 [0x5] · PMF Binary 6 [0x6] · PMF Binary 7 [0x7] · PMF Binary 8 [0x8] · PMF Binary 9 [0x9]
PMF Binary 0 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 1 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 2 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 3 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 4 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 5 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 6 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 7 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 8 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]
PMF Binary 9 Size	Selection	0 KB [0x0] ■ · 30 KB [0x1] · 60 KB [0x2] · 90 KB [0x3] · 120 KB [0x4] · 150 KB [0x5] · 180 KB [0x6] · 210 KB [0x7] · 240 KB [0x8] · 270 KB [0x9] · 300 KB [0xA]

APTS Control Method Settings (0x6E)

Submenus APTS State Index 0 ■ APTS State Index 1 ■ APTS State Index 2 ■ APTS State Index 3 ■ APTS State Index 4 ■ APTS State Index 5 ■ APTS State Index 6 ■ APTS State Index 7 ■ APTS State Index 8 ■ APTS State Index 9 ■ APTS State Index 10 ■ APTS State Index 11 ■ APTS State Index 12 ■ APTS State Index 13 ■ APTS State Index 14 ■ APTS State Index 15

Performance and Thermal State Control Method	Selection	Disabled [0x0] ■ · Enabled [0x1]

ATST Function (0x7F)

ATST Function 3	Selection	Disabled [0x0] ■ · Enabled [0x1]

APMF Function 1 Settings (0x65)

Heartbeat Time Interval	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x3C ■
PMF PPT Time Constant Control	Selection	Disabled [0x0] ■ · Enabled [0x1]
PMF PPT Time Constant	Numeric	Min 0x64 · Max 0x7530 · Step 0x1 · Default 0x64 ■

APMF Function 4 Settings (0x66)

AMPF Function 4	Selection	Disabled [0x0] ■ · Enabled [0x1]

APMF Function 7 Settings (0x67)

AMPF Function 7	Selection	Disabled [0x0] ■ · Enabled [0x1]

APMF Function 9 Settings (0x68)

AMPF Function 9	Selection	Disabled [0x0] ■ · Enabled [0x1]
APTS State Index of AC Best Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x0 ■
APTS State Index of AC Best Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x1 ■
APTS State Index of AC Best Power Efficiency	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x2 ■

APTS State Index of AC Energy Saver	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x3 ■
APTS State Index of DC Best Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x4 ■
APTS State Index of DC Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x5 ■
APTS State Index of DC Best Power Efficiency	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x6 ■
APTS State Index of DC Battery Saver	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x7 ■

APMF Function 11 Settings (0x69)

AMPF Function 11	Selection	Disabled [0x0] ■ · Enabled [0x1]
Flags	Numeric	Min 0x0 · Max 0xFFFF · Step 0x1 · Default 0x0 ■
APTS State Index of ASM Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x0 ■
APTS State Index of ASM Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x1 ■
APTS State Index of ASM Virtually Silent	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x2 ■
APTS State Index of ASM Non-Interactive	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x3 ■

APMF Function 12 Settings (0x6A)

AMPF Function 12	Selection	Disabled [0x0] ■ · Enabled [0x1]
Flags	Numeric	Min 0x0 · Max 0xFFFF · Step 0x1 · Default 0x0 ■
APTS State Index of ASM Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x4 ■
APTS State Index of ASM Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x5 ■
APTS State Index of ASM Virtually Silent	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x6 ■
APTS State Index of ASM Non-Interactive	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x7 ■

APMF Function 14 Settings (0x6B)

AMPF Function 14	Selection	Disabled [0x0] ■ · Enabled [0x1]

APMF Function 16 Settings (0x6C)

AMPF Function 16	Selection	Disabled [0x0] ■ · Enabled [0x1]

APTS State Index 0 (0x6F)

Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ ■ TUNE Selects which fan table this APTS state uses; point it at a more aggressive table for more sustained thermal headroom.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ ■ TUNE PMF power-throttler PPT cap for this APTS state. A low cap throttles the APU while this state is active; leave the default unless you tune PMF policy.
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ ■ TUNE APU-only PMF power-throttler PPT cap for this APTS state. A low cap throttles the APU while this state is active; leave the default unless you tune PMF policy.
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ ■ TUNE STT thermal/power limit. Raising the skin-temperature or min-power limit lets the APU sustain more power (more performance, hotter chassis).
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ ■ TUNE STT thermal/power limit. Raising the skin-temperature or min-power limit lets the APU sustain more power (more performance, hotter chassis).
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ ■ TUNE STT thermal/power limit. Raising the skin-temperature or min-power limit lets the APU sustain more power (more performance, hotter chassis).

APTS State Index 1 (0x70)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 2 (0x71)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 3 (0x72)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 4 (0x73)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 5 (0x74)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 6 (0x75)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 7 (0x76)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 8 (0x77)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 9 (0x78)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 10 (0x79)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 11 (0x7A)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 12 (0x7B)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 13 (0x7C)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 14 (0x7D)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

APTS State Index 15 (0x7E)

Settings identical to APTS State Index 0 (0x6F) — not reprinted.

DASH / ASF Management

1 form · 8 settings

Dash Function (0x34)

Dash Function Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Disable Multi-Function	Selection	Disabled [0x0] · Enabled [0x1] ■
Bus number Of LAN Controller	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■
Dev number Of LAN Controller	Numeric	Min 0x0 · Max 0x1F · Default 0x0 ■
Function number Of LAN Controller	Numeric	Min 0x0 · Max 0x7 · Default 0x0 ■
Asf Message Function Support	Selection	Disabled [0x0] ■ · Enabled [0x1]
Asf BIOS WatchDog Message Time Out Value	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■
Asf OS WatchDog Message Time Out Value	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■

RAIDXpert2 Configuration Utility

29 forms · 29 settings

Configuration Options (0x1)

Submenus Controller Management ■ Array Management ■ Physical Disk Management

Controller Management (0x100)

Submenus View Controller Information ■ Rescan Disks

View Controller Information (0x110)

Select Controller:	Selection	(populated at runtime)

Change Controller Properties (0x120)

Submenus Apply Changes

Convert to Legacy Disk (0x130)

Submenus OK ■ Cancel ■ Yes ■ No

Convert to Legacy Disk (0x131)

Submenus OK ■ Cancel ■ Yes ■ No

Initialize Disk(s) (0x140)

Submenus OK ■ Cancel ■ Yes ■ No

Initialize Disk(s) (0x141)

Submenus OK ■ Cancel ■ Yes ■ No

Rescan Disks (0x150)

Submenus OK

Update Controller Firmware (0x160)

Submenus Select Firmware File ■ Proceed with Firmware Download ■ Cancel Firmware Download ■ Activate Firmware ■ Cancel Firmware Activation

Firmware file extension	Selection	*.* [0x0] ■ · *.fluf [0x1] · *.fw [0x2] · *.bin [0x3] · *.raw [0x4]

Array Management (0x200)

Submenus Create Array ■ Manage Array Properties ■ Delete Array

Create Array (0x210)

Submenus Create Array ■ Select Physical Disks ■ YES ■ NO

Select RAID Level:	Selection	Volume [0x8] ■ · RAIDABLE [0xD] · RAID 0 [0x1] · RAID 1 [0x2] · RAID 5 [0x3] · RAID 6 [0xB] · RAID 10 [0x4]
Array Size:	Numeric	Min 0x0 · Max 0xFFFFFFFFFFFFFFFF · value populated at runtime
Array Size Unit:	Selection	MB (MegaBytes) [0x2] · no compiled default
Select CacheTagSize:	Selection	(populated at runtime) [0x0] ■ · 64KB [0x1] · 128KB [0x2] · 256KB [0x3]
Select CacheTagSize:	Selection	64KB [0x1] ■ · 128KB [0x2] · 256KB [0x3]
Read Cache Policy:	Selection	Read Cache [0x2] ■ · No Cache [0x1]
Write Cache Policy:	Selection	Write Back Cache [0x4] ■ · No Cache [0x3]

Select Physical Disks (0x212)

Submenus Apply Changes ■ OK ■ OK ■ Yes ■ No ■ No

Select Media Type:	Selection	HDD [0x1] ■ · SSD [0x2] · BOTH [0x3]

Select Physical Disks (0x211)

Submenus Apply Changes ■ OK ■ OK ■ Yes ■ No ■ No

Settings identical to Select Physical Disks (0x212) — not reprinted.

Select Physical Disks (0x213)

Submenus Apply Changes ■ OK ■ OK ■ Yes ■ No ■ No

Select Physical Disks (0x214)

Submenus Apply Changes ■ OK ■ OK ■ Yes ■ No ■ No

Create Array (0x215)

Submenus Create Array ■ Select Physical Disks ■ YES ■ NO

Manage Array Properties (0x220)

Submenus View Associated Physical Disks ■ Manage Dedicated Hot Spares

Select Array:	Selection	VIRT 0–31 — runtime-populated array labels [0x1]–[0x20] ■

View Associated Physical Disks (0x222)

Submenus View Physical Disk Properties

Array ID:	Text	Text input · length 0x1–0x40
RAID Level:	Text	Text input · length 0x1–0x40
Array Status:	Text	Text input · length 0x1–0x40
Array Capacity:	Text	Text input · length 0x1–0x40
Cache Tag Size:	Text	Text input · length 0x1–0x40
Hidden:	Text	Text input · length 0x1–0x40
Read Cache Policy:	Text	Text input · length 0x1–0x40
Write Cache Policy:	Text	Text input · length 0x1–0x40

Manage Dedicated Hot Spares (0x224)

Submenus Add Hot Spare Disk ■ YES ■ NO

Manage Dedicated Hot Spares (0x225)

Submenus Add Hot Spare Disk ■ YES ■ NO

Delete Array (0x230)

Submenus Delete Array(s)

Warning (0x231)

Submenus YES ■ NO

View Array Properties (0x240)

Select Array:	Selection	VIRT 0–31 — runtime-populated array labels [0x1]–[0x20] ■
Array ID:	Text	Text input · length 0x1–0x40
RAID Level:	Text	Text input · length 0x1–0x40
Array Status:	Text	Text input · length 0x1–0x40
Array Capacity:	Text	Text input · length 0x1–0x40
Read Cache Policy:	Text	Text input · length 0x1–0x40
Write Cache Policy:	Text	Text input · length 0x1–0x40

Physical Disk Management (0x300)

Submenus View Physical Disk Properties ■ Select Physical Disk Operations ■ View Global Hot Spares

View Physical Disk Properties (0x310)

Select Physical Disk:	Selection	Physical Disk 0–31: bus, size, state — runtime-populated [0x1]–[0x20] ■

Select Physical Disk Operations (0x320)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Unassign Hot Spare ■ Yes ■ No

Settings identical to View Physical Disk Properties (0x310) — not reprinted.

Select Physical Disk Operations (0x321)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Unassign Hot Spare ■ Yes ■ No

Select Physical Disk Operations (0x322)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Unassign Hot Spare ■ Yes ■ No

Appendix A — Dynamic Driver Forms (Excluded)

Contributed by generic UEFI network and boot drivers, not by this board's Setup modules. They appear only when the relevant driver binds at runtime and are not counted in the totals.

Network Stack Configuration	IPv4 / IPv6 network device policy (driver form, distinct from Setup NetworkStackSetupScreen 0x2752)	(unresolved)
IPv4 Network Configuration	Static IP, netmask, gateway, DNS (per NIC)	Ip4Dxe
IPv6 Network Configuration	IPv6 interface policy and addressing (per NIC)	Ip6Dxe
VLAN Configuration	VLAN ID and priority (per NIC)	VlanConfigDxe
HTTP Boot Configuration	UEFI HTTP/HTTPS boot URI and parameters	HttpBootDxe
TLS Auth Configuration	Server CA / client certificate enrollment	TlsAuthConfigDxe
PXE / Network Boot	CRB PXE UEFI boot driver options	CrbPxeUefiDriver

Referenced from Advanced (0x2712) in the Setup form-set but provided by other AMI drivers at runtime; not decoded, not counted.

Super IO Configuration	Super I/O device options	AMI Super I/O driver
SIO Common Setting	Super I/O common options	AMI Super I/O driver
PCI Subsystem Settings	PCI bus driver options (root-bridge and device policy)	AMI PciBus
NVMe Configuration	NVMe device information and per-drive options	AMI NVMe driver

Submenus prompts that open action or confirmation dialogs (OK / Cancel / Yes / No, Apply Changes, Decline), runtime-generated forms (UEFI BBS boot priorities, Secure Boot key management, RAIDXpert2 array and hot-spare dialogs, the controller firmware-update flow, Node / Socket information) and Setup's action-only Save & Exit page have no catalog entry.